Discrete quadrature spectrum of composite bosons: Weyl relation, intensity correlations, and binned homodyne detection

Undergraduate version

I. Roditi

CBPF – Centro Brasileiro de Pesquisas Físicas
Rua Dr. Xavier Sigaud 150, 22290-180, Rio de Janeiro, Brazil
roditi@cbpf.br

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Abstract

A composite boson made of two fermions can look elementary under one measurement and composite under another. In an exactly solvable paired-mode model we compare three probes: classical drives, periodically binned quadrature statistics, and intensity correlations $g^{(2)}$. The Weyl defect has an exact product formula and a controlled dilute expansion in the Schmidt moments. The binned observables are instead piecewise constant in the Schmidt weights and sensitive to single eigenvalues crossing bin edges; on the flat-spectrum ladder they converge slowly and nonmonotonically over the computed range. A spectral filter and a polar correction recover an exact Pauli algebra; a heralded joint projection then extracts a logical singlet from the vacua of two independent copies of the composite, with probability set by the accepted weight and the mixedness of the filtered reduced qubit. Finite-mode examples with Schmidt number about three succeed at the percent level despite order-one Weyl defects; near-unit $g^{(2)}$ does not control the binned algebra. For flat spectra the construction converges strongly to its oscillator counterpart, whose vacuum overlap is strictly positive at every scale; for infinite spectra with summable Schmidt amplitudes it converges at all but countably many filter cutoffs, positivity at the cutoffs studied remaining open.

About this version (annotated advanced-undergraduate version). This is the research paper written for a reader who has taken a first course in quantum mechanics (Dirac notation, the harmonic oscillator, spin $\frac{1}{2}$ and the Pauli matrices, the density matrix) and a course in linear algebra, but who may not have met second quantisation, the Schmidt decomposition, homodyne detection or the spectral theorem for operators. Every result, equation, number, figure and proof of the research version is kept; what is added are short *Background* paragraphs, set off by rules, that introduce each tool at the point where it is first needed, and more explicit reasoning between the equations. A reader who knows the tools can skip the ruled paragraphs without loss. A suggested route on first reading: Secs. 2–4 with their Background paragraphs (the model, the two detectors, the quadrature lattice), then Sec. 5 up to the singlet projector, then the flat-spectrum results of Sec. 6 and the conclusions; the proofs of the lemmas and propositions, the Wannier-like spectra of Sec. 7, the oscillator limit and the appendices can be read afterwards.

1 Introduction

Background: composite bosons. An electron and a hole in a semiconductor bind into an exciton; two fermionic atoms bind into a diatomic molecule; a proton and an electron make a hydrogen atom. Each of these is a pair of fermions that, as a whole, behaves like a boson: many

of them can occupy the same state, they can form Bose–Einstein condensates, and the field that creates them obeys, approximately, the commutation relation $[B, B^\dagger] = 1$ of an elementary boson such as a photon. The approximation is good when the pairs are dilute and the measurement does not resolve the constituents. Because the constituents are fermions, two pairs cannot put their electrons in the same single-particle state: this is *Pauli blocking*, and it is the source of every deviation from elementary-boson behaviour discussed in this paper.

Excitons in semiconductors and molecular crystals, Feshbach molecules of cold atoms, and biphotons are pairs of fermions or of bosons that behave as bosons when the pairs are dilute and their constituents are not resolved. How bosonic such a composite is has been quantified in several ways: by the normalisation ratios F_{N+1}/F_N of the N-pair states and the deviation of the commutator $[B, B^\dagger]$ from unity in the coboson many-body formalism [1, 2, 3, 4], by the Schmidt number or entanglement of the pair wavefunction [5, 6, 7, 8], or by the second-order correlation function $g^{(2)}$ of coherent states [3], which is also the quantity measured when exciton or polariton condensates are tested for coherence [9, 10, 11]. All of these are properties of the state or of the field algebra. A measurement, however, asks a narrower question: can *this* detector tell the composite from an elementary boson? The present paper answers that question for three detectors applied to the same composite field (Fig. 1), and finds that the answers are different.

The three detectors are the following. (A) A *classical drive*: two displacements $D_B(r)$ and $D_B(-ib)$ of the composite field with $2rb = \pi$, which for an elementary boson anticommute exactly (the Weyl relation, explained in Sec. 3). (B) *Homodyne detection with periodically binned outcomes*: the quadrature outcome q (a continuous number, explained in Sec. 4) is recorded only through the sign of $\cos q$, which converts the two conjugate quadratures into two-valued observables $X = \operatorname{sgn}\cos Q_r$ and $Z = \operatorname{sgn}\cos Q_w$. For an elementary boson these two observables anticommute exactly, so that two independent oscillators – one held by each of two parties, Alice and Bob – carry commuting Pauli algebras, and their joint vacuum $|0\rangle_A|0\rangle_B$ (the "two vacua" referred to throughout) has a nonzero overlap with the spin singlet of the two Pauli pairs [12]. Dichotomised homodyne outcomes have a long history in continuous-variable Bell tests, from sign binning [13, 14, 15, 16] to parity and pseudospin operators [17, 18]; periodic binning is the modular-variable construction [19, 20, 21], and the same coarse graining underlies the mutually unbiased coarse-grained measurements of Refs. [22, 23]; Bell–CHSH tests built from Weyl operators and squeezed states have also been formulated directly for bosonic quantum fields [24]. In all of that work the field is an elementary boson; here it is a composite. (C) *Intensity correlations*: the normally ordered moment $g^{(2)} = \langle B^{\dagger 2} B^2\rangle / \langle B^\dagger B\rangle^2$ measured by photon or atom counting, which equals 1 for a coherent (laser-like) field and 2 for a thermal one. Of the three, detector (B) is the one that matters if the composite is to be used as a qubit resource, and it is the one for which bosonicity has not been examined.

Background: why anticommutation matters. The Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ satisfy $\sigma_x^2 = \sigma_z^2 = I$ and $\sigma_x\sigma_z = -\sigma_z\sigma_x$: they *anticommute*. Conversely, any two Hermitian operators X, Z with $X^2 = Z^2 = I$ and $XZ = -ZX$ generate a copy of the Pauli algebra (with $Y = iXZ$), possibly many copies at once. This is what makes a qubit: once one has such a pair of observables on each of two systems, one can write down the projector onto the spin singlet, $\frac{1}{4}(I - X\otimes X - Y\otimes Y - Z\otimes Z)$, and test Bell inequalities with them. If X and Z only *approximately* anticommute, that expression is not a projector and nothing that follows is exact. The central technical object of this paper is therefore the *anticommutator* $\{X, Z\} = XZ + ZX$ of two binned quadratures, which should be zero and, for a composite boson, is not.

Detector (B) is not an arbitrary choice. Sign-binned quadratures at the Weyl area $2rb = \pi$ are the periodic coarse graining of the modular-variable constructions cited above, and the Gottesman–Kitaev–Preskill encoding [25] is built on the same structure: its logical Pauli operators are displacements, and its logical measurement is a homodyne measurement whose outcome is decoded periodically. The globally anticommuting binned quadratures studied here are one realisation of a qubit algebra in a continuous-variable field – not the only one, since a qubit can also be encoded

in two oscillator states – but it is the one that these constructions use, and all of them have been formulated for elementary bosons. If composite fields – exciton-polaritons, molecular condensates, or the collective excitations of spin ensembles – are to serve in such protocols, the relevant question is whether *their* binned quadratures anticommute, and the conventional bosonicity criteria do not answer it: Secs. 6 and 8 show that a Weyl defect of 10^{-3} coexists with a binned defect of 0.37, and $g^{(2)} = 0.99999$ with a binned defect of order one. The question has a positive side as well: Sec. 5 shows that a composite whose binned quadratures fail to anticommute can still yield an exact logical qubit, with the rejected weight computable from the spectrum.

Where such measurements exist. For a flat spectrum the composite field is a collective spin, $B = J_-/\sqrt{S}$ (Sec. 2), and the two binned quadratures are coarse-grained measurements of J_x and J_y on an ensemble of S two-level systems – the Holstein–Primakoff setting [26], in which collective-spin measurements are routine [27]. Our flat-spectrum results are therefore exact statements about when the bosonic approximation of a finite ensemble survives a sign-binned measurement. For exciton-polaritons the quadratures of the emitted light are measured by homodyne detection [10], although relating them to the composite field requires a coupling model (Sec. 2); for trapped dimers the coboson formalism applies directly [28], but a quadrature measurement of a molecular field has no established implementation. In every case the binned measurement changes the pair number and therefore presupposes a phase reference or reservoir (Sec. 5.2), as does any displacement.

At finite Schmidt rank, the quadrature of the composite has a *discrete spectrum*, which makes its response to binning sensitive to individual eigenvalues. If $B^\dagger = \sum_k \sqrt{\lambda_k}\, a_k^\dagger b_k^\dagger$ is the Schmidt decomposition of the pair (Sec. 2), the eigenvalues of a quadrature are the signed sums $\sum_k \pm\sqrt{\lambda_k}$: 2^S values counted with multiplicity for S Schmidt modes, which for a flat spectrum reduce to $S+1$ equally spaced points with spacing $\propto 1/\sqrt{S}$, whereas for the infinite geometric family the support is an interval or a Cantor set depending on the decay parameter. For both infinite families considered below, the quadrature spectral measures have no atoms[1] (Sec. 7). An elementary boson's quadrature is continuous. The probes respond differently to this spectral structure. A drive is a smooth probe – the fundamental harmonic of the quadrature – so its defect is a real-analytic function of the Schmidt weights, computable in closed form, with a dilute expansion controlled by the first Schmidt moments. Intensity correlations do not involve the quadrature at all. Binned homodyne, with bins of fixed width against a lattice whose spacing shrinks with S, resolves the lattice point by point: its operators change only when a lattice point crosses a bin edge, they are therefore piecewise constant in the Schmidt weights, the purity and the third moment do not determine them, and their convergence to the oscillator algebra is slow and nonmonotone. Pauli blocking thus has two faces – the commutator, seen by the drive, and the discreteness of the quadrature spectrum, seen by binned homodyne – and neither the moment expansions of coboson theory nor $g^{(2)}$ bridges them.

Two conclusions follow. First, bosonic quality is not what the qubit task requires: a composite with Schmidt number 3.00 (a 19-mode truncation of a geometric spectrum), whose normalisation ratio $F_2/F_1 = 1 - P$ is a third below unity and whose Weyl defect is of order one, still yields an exact qubit singlet from two vacua with probability 1.2%, after a spectral filter and a polar correction that we specify. Second, the singlet can be extracted from two vacua only because, in the factorisation that the two observables define, the qubit is entangled with its complementary logical factor – a structure that the elementary oscillator shares and that compositeness modulates: the success probability is the squared accepted weight times the mixedness of the filtered reduced qubit, a function of the Schmidt spectrum which vanishes at many small ranks and vanishes identically when the spectrum is so concentrated that $\sum_k \sqrt{\lambda_k} < \pi/2$ at $r = 1$, for then one binned quadrature is the identity.

Section 2 defines the model and physical scales. Sections 3 and 4 compare the smooth dis-

[1] An *atom* of a measure is a single point carrying positive measure. The spectral measure of a quadrature Q in a state ψ assigns to each set of real numbers the probability that a measurement of Q in ψ gives a value in that set; it has no atoms when no single value has positive probability. The bin edges of sgn cos then carry no weight, which is the property the convergence arguments of Sec. 7 use. A Background paragraph in Sec. 7 says more.

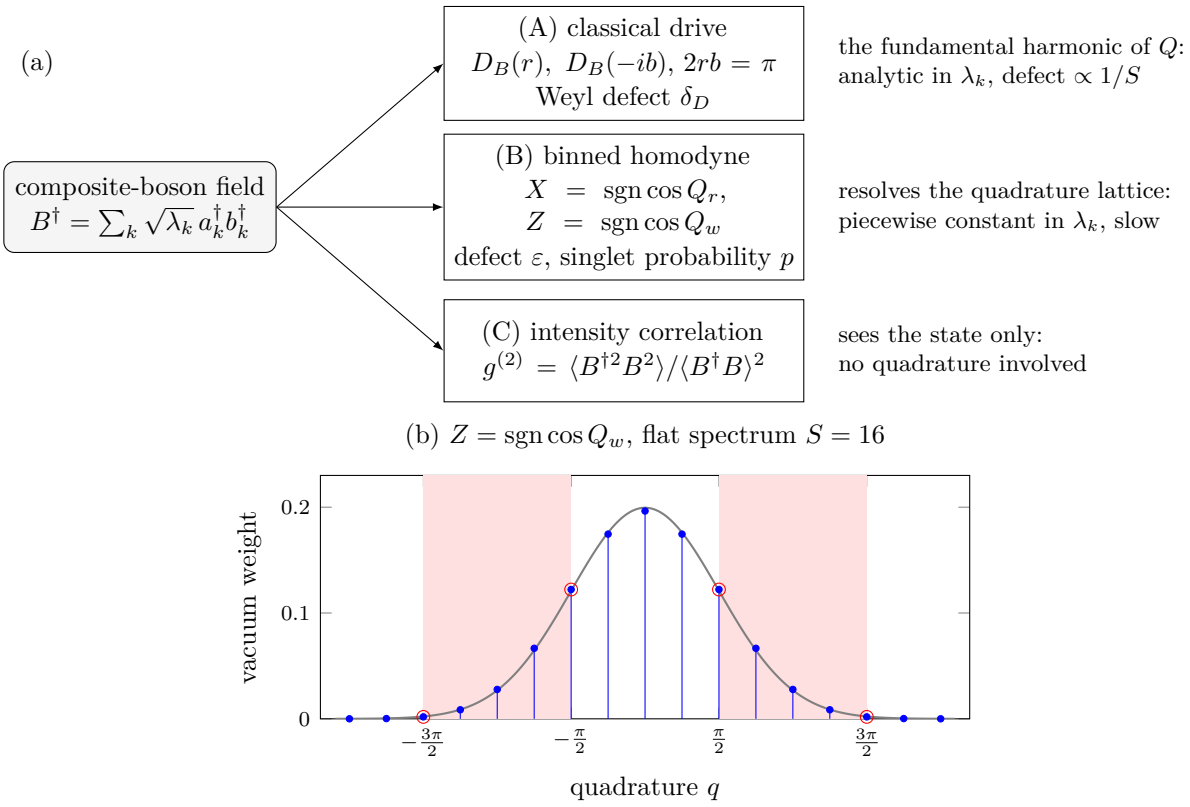


Figure 1: (a) The three detectors applied to one composite-boson field, and what each of them resolves of the quadrature spectrum. (b) The discrete quadrature spectrum behind the results: eigenvalues of $Q_w = -(\pi/2)(B + B^\dagger)$ for a Frenkel-like coboson with $S = 16$ sites (stems, with the vacuum weight on each), against the bins of the sign function (shaded: $Z = -1$) and the oscillator vacuum distribution (grey line, scaled to the lattice spacing). The lattice has spacing $\pi/\sqrt{S}$ against bins of width π; here four lattice points fall exactly on bin edges (circles), carrying 25% of the vacuum weight, so the sign of Z on them is a convention and the binned operator at this rank depends on it (Sec. 6). A drive responds to this distribution through its characteristic function at one frequency, a smooth functional dominated in the dilute regime by the first moments; binned homodyne sees every point.

placement defect with the lattice-sensitive binned observables. Section 5 applies the standard two-involution reduction [29, 30] to recover an exact Pauli algebra and derives the preparation probability and correction bounds. Sections 6 and 7 treat flat and nonflat spectra; Sec. 8 compares intensity correlations. Section 9 proves the flat-spectrum oscillator limit and the positivity of its vacuum overlap. Section 10 discusses the conclusions and open problems, and Appendix C corrects a published Anderson-state correlation. The hard-bin convergence rate remains open. At fixed nonzero limiting purity, convergence holds for all but countably many cutoffs; convergence and positive limiting probability at the numerical cutoffs remain unproved.

Preparation includes a joint entangling operation. A subsequent Bell violation therefore concerns the heralded prepared state, rather than the initial product vacuum. Its interpretation as a Bell test requires preparation to be completed and heralded before independently chosen local measurement settings (Sec. 5.2). The paper analyses the relation between the constituent algebra, the measurements and this specified preparation protocol; an experimental implementation remains to be developed.

2 Paired-mode model and the Frenkel ladder

Background: creation operators and the Schmidt decomposition. In second quantisation a state with one particle in orbital k is written $a_k^\dagger|0\rangle$, where $|0\rangle$ is the vacuum (no particles) and $a_k^\dagger$ is the *creation operator* for that orbital; a_k removes the particle. For fermions $a_k^\dagger a_k^\dagger = 0$ (no two fermions in one orbital) and operators of different orbitals anticommute. A pair of distinguishable fermions – an electron (a) and a hole (b) – in a bound state has a two-body wavefunction $\phi(x_a, x_b)$. The *Schmidt decomposition* is the statement, true for any such function, that one can choose orthonormal orbitals $\{u_k\}$ for the electron and $\{v_k\}$ for the hole such that $\phi(x_a, x_b) = \sum_k \sqrt{\lambda_k}\, u_k(x_a) v_k(x_b)$ with $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$ (it is the singular value decomposition of ϕ regarded as a matrix). The number of nonzero λ_k is the Schmidt rank S; the weights λ_k measure how strongly the two particles are correlated. If only one λ_k is nonzero the wavefunction is a product and the particles are uncorrelated; the more evenly the weight is spread, the more entangled they are. The *purity* $P = \sum_k \lambda_k^2$ and the *Schmidt number* $\mathcal{K} = 1/P$ summarise this: $\mathcal{K} = 1$ for a product, $\mathcal{K} = S$ for a flat spectrum $\lambda_k = 1/S$.

We consider two distinguishable fermion species and one normalised pair wavefunction of finite Schmidt rank S. After absorbing fixed phases into the constituent orbitals, the pair creation operator reads

$$B^\dagger = \sum_{k=1}^{S} \sqrt{\lambda_k}\, d_k^\dagger, \qquad d_k^\dagger = a_k^\dagger b_k^\dagger, \qquad \lambda_k > 0, \quad \sum_k \lambda_k = 1. \tag{1}$$

The operator $d_k^\dagger$ creates the electron in orbital k *and* the hole in orbital k; the pair operator $B^\dagger$ is a weighted sum of these. Because the Schmidt orbitals are orthonormal, the states in which the pair modes k are either empty or occupied form an invariant subspace – the *paired sector* – with the local basis $|0\rangle_k$, $|1\rangle_k = d_k^\dagger|0\rangle_k$ for each mode. Distinct pair operators commute (each contains two fermion operators, so the signs cancel), $(d_k^\dagger)^2 = 0$ (Pauli blocking: a second pair cannot use the same orbitals) and $[d_k, d_k^\dagger] = I - 2n_k$, where n_k is the occupation of mode k. Hence the sector is isomorphic to S two-level systems, and

$$[B, B^\dagger] = I - \Delta, \qquad \Delta = 2\sum_k \lambda_k n_k, \qquad [\Delta, B^\dagger] = 2\sum_k \lambda_k^{3/2} d_k^\dagger. \tag{2}$$

The first relation is the heart of the matter. For an elementary boson $[B, B^\dagger] = I$; for the composite the commutator is I minus the *deviation operator* Δ, which counts the occupied modes weighted by their Schmidt weights. It vanishes on the vacuum, $\Delta|0\rangle = 0$, but it does not vanish on states with pairs present, and the second relation shows that commuting Δ with $B^\dagger$ produces yet another operator. This matters as soon as one exponentiates: combining two finite rotations generated by $B^\dagger - B$ and $B^\dagger + B$ brings in the commutator $[B, B^\dagger] = I - \Delta$ and then its commutators with the generators, $[\Delta, B^\dagger] = 2\sum_k \lambda_k^{3/2} d_k^\dagger$ and so on, which do not vanish on the vacuum ($[\Delta, B^\dagger]|0\rangle \neq 0$) and are not numbers. (Individual nested commutators such as $[B, [B, B^\dagger]] = -2\sum_k \lambda_k^{3/2} d_k$ do annihilate the vacuum; it is the full series that fails to collapse to a phase.) This is why the Weyl relation of Sec. 3 is only approximate. Interactions and singly occupied modes are outside the model. In the two-party setting below, Alice and Bob are assigned disjoint constituent supports, so that their local algebras commute.

Background: displacement operators and quadratures. For an elementary boson with $[a, a^\dagger] = 1$, the displacement operator $D(z) = e^{za^\dagger - z^* a}$ is the unitary that a classical drive (a laser pulse, say) applies to the field: it maps the vacuum to the coherent state $|z\rangle$. It can be written as e^{iQ} with $Q = -i(za^\dagger - z^* a)$ Hermitian. With $\hat{x} = (a + a^\dagger)/\sqrt{2}$ and $\hat{p} = -i(a - a^\dagger)/\sqrt{2}$, so that $[\hat{x}, \hat{p}] = i$: for real $z = r$, $Q_r = -ir(a^\dagger - a) = -r\sqrt{2}\,\hat{p}$ is proportional to the "momentum" quadrature; for imaginary $z = -ib$, $Q = -b(a^\dagger + a) = -b\sqrt{2}\,\hat{x}$ is proportional to the "position" quadrature. Homodyne detection measures a quadrature of an optical field by interfering it with a strong reference beam; the outcome is a real number q with a continuous distribution. In the vacuum, $\hat{x}$ and $\hat{p}$ are Gaussian with variance $\frac{1}{2}$, so the outcomes of Q_r and Q_w are Gaussian with variances r^2 and b^2.

The unitary displacements and their Hermitian generators are

$$D_B(z) = e^{zB^\dagger - z^*B} = e^{iQ_z}, \qquad Q_z = -i\,(zB^\dagger - z^*B), \qquad z = r, \quad w = -ib, \quad b = \frac{\pi}{2r}, \quad r > 0, \tag{3}$$

i.e. $Q_r = -ir(B^\dagger - B)$ and $Q_w = -b(B^\dagger + B)$, with

$$[Q_r, Q_w] = -i\pi\,(I - \Delta). \tag{4}$$

For an elementary boson this commutator would be the number $-i\pi$. The canonical oscillator is obtained when the deviation operator disappears in an appropriate limit, not by setting it to zero at finite rank.

Flat spectrum.

Background: Dicke states and the collective spin. Take S two-level systems ("spins") and the states that are symmetric under permutation of the spins. There are $S+1$ of them, labelled by the number N of spins pointing up; they are the Dicke states, and on them the collective operators $J_\pm = \sum_k \sigma_\pm^{(k)}$, $J_z = \frac{1}{2}\sum_k \sigma_z^{(k)}$ act exactly as the angular-momentum operators of a single spin of magnitude $S/2$. If every Schmidt weight is the same, $\lambda_k = 1/S$, then $B^\dagger = S^{-1/2}\sum_k d_k^\dagger$ is symmetric in the modes and generates Dicke states from the vacuum.

For $\lambda_k = 1/S$ the states generated from the vacuum are Dicke states and the ladder closes exactly,

$$\begin{aligned} \|(B^\dagger)^N|0\rangle\|^2 &= N!\,F_N, \qquad F_N = \frac{S!}{S^N(S-N)!}, \\ B^\dagger|N\rangle &= \sqrt{(N+1)\Big(1-\tfrac{N}{S}\Big)}\,|N+1\rangle, \qquad B|N\rangle = \sqrt{N\Big(1-\tfrac{N-1}{S}\Big)}\,|N-1\rangle, \end{aligned} \tag{5}$$

for $0 \le N \le S$; these are the Frenkel-like relations of Ref. [3] with $S = N_s$ (a Frenkel exciton is one delocalised over N_s molecular sites, with the same amplitude on each). Compare with an elementary boson, for which $B^\dagger|N\rangle = \sqrt{N+1}|N+1\rangle$: the factor $(1-N/S)$ is Pauli blocking, which makes it harder to add the $(N+1)$th pair when N of the S available orbitals are already used, and which forbids an $(S+1)$th pair altogether. The normalisation factor $F_N = \prod_{k=0}^{N-1}(1-k/S)$ satisfies $\log F_N = -N(N-1)/(2S) + O(N^3/S^2)$ for $N/S \to 0$, reaches $F_S = S!/S^S$, and the ladder terminates at $N = S$, $B^\dagger|S\rangle = 0$: this is the moth-eaten effect of Refs. [31, 3], and it is contained in every flat-spectrum calculation below through the coefficients in (5). The orthogonal annihilation remainder vanishes identically. Equivalently, $B = J_-/\sqrt{S}$ for a spin $S/2$, $B + B^\dagger = 2J_x/\sqrt{S}$ and $-i(B^\dagger - B) = 2J_y/\sqrt{S}$: every flat-spectrum result below is a statement about functions of collective spin components of a spin $S/2$. This is the Holstein–Primakoff setting [26]: the oscillator limit of Sec. 9 is the bosonic approximation of the ensemble, and the binned quadratures of Sec. 4 are sign functions of J_x and J_y with bin widths $\Delta m_x = r\sqrt{S}$ and $\Delta m_y = \pi\sqrt{S}/(2r)$. At $r = 1$ these are respectively 2 and π times the projection noise $\sqrt{S}/2$ of the polarised state $|J, -J\rangle$, so the flat-spectrum results are exact statements about a finite spin ensemble under such a measurement [27]. We keep the coboson language because it fixes the normalisation and because the nonflat case (Sec. 7) is not a single spin. The pair-counting operator $\hat{N}$ differs from the field intensity,

$$B^\dagger B = \hat{N}\Big(1 - \frac{\hat{N}-1}{S}\Big), \qquad [B, B^\dagger] = I - \frac{2\hat{N}}{S}, \tag{6}$$

so the saturated state contains S pairs but has intensity $\langle B^\dagger B\rangle = 1$; this is the saturation side of the moth-eaten effect.

Physical scales. The Schmidt rank and purity $P = \sum_k \lambda_k^2$ (Schmidt number $\mathcal{K} = 1/P$ [5]) depend on the pair wavefunction and its confinement. A Frenkel exciton delocalised uniformly over N_s molecular sites gives a flat spectrum with $S = N_s$ [3]. Wannier excitons and trapped dimers – to which the coboson formalism has been applied directly for strongly bound fermion pairs [28] – motivate nonflat spectra, but their Schmidt weights require a microscopic wavefunction; a finite numerical cutoff S is not itself a measured coherence length or an inverse purity. Rough estimates nevertheless indicate the range: the exciton coherence length in molecular aggregates is a few to of order 10^2 sites, and for a Wannier exciton the Schmidt number grows with the ratio of confinement size to Bohr radius (cf. the trapped-pair estimates of Ref. [5]), from near 1 in the strong-confinement limit, where electron and hole are confined separately and the pair wavefunction factorises, to of order 10 in a dot a few Bohr radii across and to large values in bulk. In polariton condensates the composite nature of the exciton also becomes visible at high density, where a BCS-like description takes over from the bosonic one [32]. The calculations below therefore compare two model regimes: increasing flat rank, where the oscillator limit emerges, and increasing truncation rank at fixed nonzero limiting purity. Assigning either regime to a particular aggregate, quantum dot or atomic trap requires a system-specific Schmidt decomposition. Likewise, inferring composite quadrature statistics from emitted polariton light [10] requires a coupling and detection model that we do not supply; the correlations of microcavity emission are themselves shaped by the detection optics [33].

3 Detector (A): the response to classical drives

Background: the Weyl relation. For an elementary boson, $D(z)D(w) = e^{(zw^*-z^*w)/2}D(z+w)$, and therefore $D(z)D(w) = e^{zw^*-z^*w}D(w)D(z)$: two displacements commute up to a phase that depends only on the "area" zw^*-z^*w they span in the complex plane. This is the Weyl form of the canonical commutation relation; it follows from the Baker–Campbell–Hausdorff formula because the commutator of the generators is a number. For $z = r$ and $w = -ib$ the phase is e^{2irb}, and with $2rb = \pi$ it equals -1: the two displacements *anticommute*, $D(r)D(-ib) = -D(-ib)D(r)$. For the composite the commutator of the generators is an operator, the phase is not a number, and the relation fails by an amount we now compute.

A classical drive displaces the composite field, and two drives with $2rb = \pi$ probe the Weyl relation. For an elementary boson $D(r)D(-ib) = -D(-ib)D(r)$; for the composite the failure of this relation on the vacuum is a function of the Schmidt spectrum that we now compute exactly. The local generators in (3) commute between different k, so the vacuum characteristic function factorises,

$$\phi_z(t) = \langle 0|e^{itQ_z}|0\rangle = \prod_k \cos\left(t|z|\sqrt{\lambda_k}\right). \tag{7}$$

(Each mode k contributes a two-level system on which Q_z acts as $|z|\sqrt{\lambda_k}$ times a Pauli matrix, whose exponential is cos plus i sin times the Pauli matrix; the vacuum expectation keeps only the cosine.) Read as a distribution, Eq. (7) says that the vacuum outcomes of Q_z are the signed sums $|z|\sum_k \varepsilon_k\sqrt{\lambda_k}$ with independent fair signs ε_k; a drive, $D_B(z) = e^{iQ_z}$, is the value of this characteristic function at $t = 1$, which is the precise sense in which it probes the quadrature through its fundamental harmonic only. Define the displacement anticommutation defect for the choice $2rb = \pi$,

$$\delta_D = \left\| \left[D_B(r)D_B(-ib) + D_B(-ib)D_B(r)\right]|0\rangle \right\|. \tag{8}$$

This is the norm of the vector that should be zero if the two displacements anticommuted on the vacuum; it is 0 for an elementary boson and at most 2. On a single pair mode the two displacements are the rotations

$$U_k = \begin{pmatrix} \cos\alpha_k & -\sin\alpha_k \\ \sin\alpha_k & \cos\alpha_k \end{pmatrix}, \qquad V_k = \begin{pmatrix} \cos\beta_k & -i\sin\beta_k \\ -i\sin\beta_k & \cos\beta_k \end{pmatrix}, \qquad \alpha_k = r\sqrt{\lambda_k}, \quad \beta_k = b\sqrt{\lambda_k}, \tag{9}$$

with local group-commutator expectation

$$g_k = \langle 0|V_k^\dagger U_k^\dagger V_k U_k|0\rangle = 1 - 2\sin^2\alpha_k \sin^2\beta_k - \frac{i}{2}\sin 2\alpha_k \, \sin 2\beta_k. \tag{10}$$

Expanding the squared norm in (8) – it equals $2+2\,\mathrm{Re}\langle 0|(D_r D_w)^\dagger D_w D_r|0\rangle$ – and using the product structure gives the exact expression

$$\delta_D^2 = 2 + 2\,\mathrm{Re}\prod_k g_k, \tag{11}$$

without any truncation of the Baker–Campbell–Hausdorff series. Being a product of entire functions of $\sqrt{\lambda_k}$, δ_D^2 is a real-analytic function of the Schmidt weights: it can be expanded in a convergent power series around any spectrum, and in particular it has no jumps.

Background: moments of the Schmidt spectrum. Coboson theory expands many-body quantities in the "Schmidt moments" $M_j = \sum_k \lambda_k^j$, of which $M_2 = P$ is the purity. For a flat spectrum $M_j = S^{1-j}$, so each successive moment is smaller by $1/S$; for a general spectrum the moments decrease with the largest weight $\lambda_{\max}$, since $M_{j+1} \le \lambda_{\max} M_j$. An expansion in moments is thus an expansion in the "dilution" of the composite.

Let $P = \sum_k \lambda_k^2$, $M_j = \sum_k \lambda_k^j$ and $L = r^2 + b^2$. On the branch of the logarithm continued from $\lambda = 0$,

$$\begin{gathered}\log g(\lambda) = -2irb\,\lambda + iA\lambda^2 - (D + iE_3)\lambda^3 + O(\lambda^4),\\ A = \tfrac{4}{3}rbL, \qquad D = 2r^2b^2L, \qquad E_3 = \frac{4rb(r^4+b^4)}{15} + \frac{20r^3b^3}{9}.\end{gathered} \tag{12}$$

Summed over modes with $\sum_k \lambda_k = 1$, the first-order term gives $e^{-2irb} = -1$, the elementary-boson value of the group commutator; the defect is built entirely from the remaining orders, which is why it starts at P^2 and M_3 rather than at P. For fixed r, b and sufficiently small largest weight $\lambda_{\max}$, analyticity bounds the summed remainder by a constant times M_4, and the expansion of $2 - 2\,\mathrm{Re}\,e^X$ with $X = iAP - (D + iE_3)M_3 + O(M_4)$ gives

$$\delta_D^2 = A^2P^2 + 2DM_3 + O\big(M_4 + PM_3 + P^4\big). \tag{13}$$

(The first-order terms $-2irb\lambda_k$ sum to $-i\pi$, whose exponential is -1; this is the Weyl phase. The second-order terms sum to iAP, a pure phase error; the third-order terms contribute both a phase and a loss of modulus DM_3. Squaring and adding gives the displayed terms.) The remainder is smaller than the displayed terms as $\lambda_{\max} \to 0$, by $P^2 \le M_3$, $M_4 \le \lambda_{\max} M_3$ and $P \le \lambda_{\max}$. The third moment is therefore relevant at leading order for a general spectrum. For the flat spectrum,

$$\delta_D = \frac{\pi\sqrt{L + 4L^2/9}}{S} + O(S^{-2}), \tag{14}$$

with coefficient $9.3252393\ldots$ at $r = 1$ and minimum $8.6196987\ldots$ at the balanced scale $r = b = \sqrt{\pi/2}$. This optimisation concerns detector (A) only: at $S = 8$ the balanced scale gives a vanishing preparation probability for detector (B) (Sec. 6). The physical content of this section is that a drive, being a smooth (fundamental-harmonic) probe of the quadrature, responds to the composite through a real-analytic function of the spectrum, whose dilute expansion is governed by the first Schmidt moments; everything about detector (A) is smooth in the spectrum.

4 Detector (B): binned homodyne and the quadrature lattice

Background: functions of an operator. A Hermitian operator Q has real eigenvalues q_j and orthonormal eigenvectors $|q_j\rangle$, and any function F of a real variable defines an operator $F(Q) = \sum_j F(q_j)|q_j\rangle\langle q_j|$: one applies F to the eigenvalues and keeps the eigenvectors. (For

operators with continuous spectrum, such as the position of a particle, the sum becomes an integral and the statement is the spectral theorem.) Measuring $F(Q)$ gives the same outcome statistics as measuring Q and then applying F to the outcome; the two procedures differ in what they do to the state, since a projective measurement of $F(Q)$ projects onto the whole eigenspace of one sign (a coherent superposition of many Q eigenvectors), whereas measuring Q first collapses onto a single eigenvector of Q. The constructions of Sec. 5 use the former. Here $F(q) = \operatorname{sgn}\cos q$ takes only the values ± 1, so $F(Q)$ is a two-valued observable with $F(Q)^2 = I$: an *involution*. The eigenvectors of $F(Q)$ are those of Q, grouped by the sign of $\cos q_j$.

Homodyne detection of the composite field yields a quadrature outcome q; recording only the sign of $\cos q$ coarse-grains it into bins of width π with alternating signs. This defines the outcome statistics of the local observables; the preparation protocols of Sec. 5 and Appendix A assume in addition coherent (Lüders) projections[2] onto the joint eigenspaces of two-party observables such as $Z \otimes Z$, which ordinary homodyne detection followed by classical binning does not implement. We use the explicit bin convention

$$X = F(Q_r), \qquad Z = F(Q_w), \qquad F(q) = \begin{cases} +1, & \cos q \geq 0, \\ -1, & \cos q < 0, \end{cases} \qquad \varepsilon_v = \|(XZ + ZX)v\|. \tag{15}$$

Both are self-adjoint involutions. For the oscillator they anticommute exactly [12]; for the coboson they do not, and ε_v measures the failure on a normalised initial state v. The difference is the spectrum: the oscillator quadrature is continuous, whereas Q_w has the 2^S eigenvalues $b\sum_k (-1)^{t_k}\sqrt{\lambda_k}$, counted with multiplicity – the discrete spectrum of Fig. 1(b). To see where these eigenvalues come from, note that $Q_w = -b\sum_k \sqrt{\lambda_k}(d_k^\dagger + d_k)$, and that on mode k the operator $d_k^\dagger + d_k$ is the Pauli matrix σ_x in the basis $|0\rangle_k, |1\rangle_k$, with eigenvalues ± 1 and eigenvectors $(|0\rangle_k \pm |1\rangle_k)/\sqrt{2}$. Since the modes commute, the eigenvectors of Q_w are the 2^S products of these, labelled by a string of signs $t = (t_1, \ldots, t_S)$, and the eigenvalues are the corresponding signed sums. The same holds for Q_r with σ_y in place of σ_x. Three structural facts follow from this and govern everything below.

Parity. F is even and Q_r, Q_w are odd under the pair-number parity $(-1)^{\hat{N}}$ (they change the number of pairs by one), so X and Z commute with $(-1)^{\hat{N}}$ and preserve number parity. For the flat spectrum the even and odd Dicke sectors can be treated separately; the quadrature $B + B^\dagger$ has eigenvalues $(S - 2j)/\sqrt{S}$, $j = 0, \ldots, S$ (there are j minus signs among the S), and X, Z are obtained exactly by diagonalising a tridiagonal $(S+1) \times (S+1)$ matrix (a Krawtchouk, or Dicke-basis, spectral transform). This is a reduction to an invariant physical subspace, not a numerical oscillator truncation.

Piecewise constancy and low moments. At fixed rank the spectrum of Q_w consists of signed sums bq_t, where $q_t = \sum_k (-1)^{t_k}\sqrt{\lambda_k}$. Its product eigenvectors do not depend on the weights; those of Q_r differ by a fixed number-dependent phase. Hence X and Z are constant within each region of fixed bin signs, and change only at bin crossings: if one varies the λ_k continuously, nothing happens to Z until some bq_t reaches a value $\pm\pi/2, \pm 3\pi/2, \ldots$, at which point one eigenvalue of Z flips from $+1$ to -1 and Z jumps. Expectation values and preparation probabilities inherit this property when the initial vector and other protocol parameters are fixed. A weight-dependent initial state need not do so.

Purity and the third moment do not determine the binned observables. The rank-four spectra

$$\lambda^{(a)} = (0.49,\ 0.01,\ 0.25,\ 0.25), \qquad \lambda^{(b)} = (0.442,\ 0.058,\ 0.394,\ 0.106) \tag{16}$$

have $P = 913/2500$ and $M_3 = 1489/10000$, but at $r = 1$ their squared vacuum defects are 1 and $5/2$, respectively. The difference is visible in the bins: for $\lambda^{(a)}$ the two drives cut the set of signed

[2] The state update $\rho \mapsto P\rho P/\mathrm{Tr}(P\rho P)$ for the projector P onto the selected joint eigenspace, which keeps the coherence within that eigenspace. Recording the outcome of a local homodyne measurement and binning it classically produces the same outcome statistics but not this state.

sums at the same places – the signs $F(rq_t)$ and $F(bq_t)$ coincide for every t, both being -1 exactly at $|q_t| = 1.6$ and 1.8 – whereas for $\lambda^{(b)}$ they cut it at different ones ($F(rq_t) = -1$ only at $|q_t| = 1.86$, $F(bq_t) = -1$ at $|q_t| = 1.21$, 1.38 and 1.86). A specified sequential preparation also gives different probabilities and Bell values (Appendix A). These spectra are outside the dilute regime of (13); they establish nondetermination by these two moments, not a failure of that expansion. They do not show that moments cannot give bounds. Indeed, at known finite rank, the power sums through order S reconstruct the unordered spectrum by Newton's identities (the first S power sums of S numbers determine the numbers up to order), and determine the vacuum protocol up to mode relabeling.

High harmonics and edge projectors.

Background: Fourier series of the square wave. The square wave $F(q) = \operatorname{sgn}\cos q$ is periodic with period 2π and has the Fourier series $\frac{4}{\pi}(\cos q - \frac{1}{3}\cos 3q + \frac{1}{5}\cos 5q - \dots)$. The coefficients decay only as $1/(2m+1)$ because of the jumps; the partial sums overshoot near the jumps (the Gibbs phenomenon), and at a jump itself they converge to the midpoint, here 0. The *Fejér* sums – the averages of the first M partial sums – avoid the overshoot: they are bounded by 1 and converge to $F(q)$ at every point where F is continuous, and to 0 at the jumps.

Away from its discontinuities the square wave has the Fourier series

$$F(q) = \frac{4}{\pi}\sum_{m\geq 0}\frac{(-1)^m}{2m+1}\cos(2m+1)q.$$

At each bin edge $c \in \mathcal{C} = \pi/2 + \pi\mathbb{Z}$, every term vanishes, whereas our convention assigns $F(c) = +1$. If F_M^{F} denotes the Fejér approximation, the spectral theorem gives, with s-lim denoting the strong operator limit (convergence on every vector),

$$F(Q) = \operatorname*{s\text{-}lim}_{M\to\infty} F_M^{\mathrm{F}}(Q) + P_{\mathrm{edge}}(Q), \qquad P_{\mathrm{edge}}(Q) = \mathbf{1}_{\mathcal{C}}(Q). \tag{17}$$

where $\mathbf{1}_{\mathcal{C}}(Q)$ is the projector onto the eigenvectors of Q whose eigenvalue lies exactly on a bin edge, and "s-lim" means that the limit holds when both sides are applied to any fixed vector. The last term has a minus sign for the alternative edge convention. It vanishes for the oscillator spectral measure (a continuous distribution gives zero probability to the countable set $\mathcal{C}$), but generally not at finite rank, where a lattice point may sit exactly on an edge. With $X = \bar{X} + P_r$, $Z = \bar{Z} + P_w$, where the bars denote the Fejér limits, the anticommutator includes

$$\{X, Z\} = \{\bar{X}, \bar{Z}\} + \{\bar{X}, P_w\} + \{P_r, \bar{Z}\} + \{P_r, P_w\}. \tag{18}$$

A Fourier-only estimate cannot omit these terms at a finite-rank edge crossing. Even without edge atoms, low-order displacement estimates are not uniform over all harmonics: the mth harmonic $\cos(2m+1)Q_r$ is a displacement by $(2m+1)r$, so to control the whole square wave one needs the Weyl relation for displacements of all sizes, whereas Sec. 3 controls it only for small ones. Multiplying the limiting Gaussian by a position square wave also creates jumps and takes the vector outside the kinetic-energy form domain (its derivative is not square-integrable), so finite-occupation error estimates cannot be propagated through the measurement sequence without further control.

4.1 Stability and a smooth measurement model

Let $\gamma_a = \min_t \operatorname{dist}(aq_t, \mathcal{C})$ for $a = r, b$: the distance from the closest lattice point to the closest bin edge. These margins belong to the finite spectrum; $0 \leq \gamma_a \leq \pi/2$.

Proposition 1. *Consider two spectra of the same rank in the same paired basis, with scales* (r, b) *and* (r', b')*. Put*

$$d = \sum_k |\sqrt{\lambda'_k} - \sqrt{\lambda_k}|, \qquad \zeta_a = a'd + |a' - a|\sum_k \sqrt{\lambda_k}, \quad a \in \{r, b\}. \tag{19}$$

If $\zeta_r < \gamma_r$ and $\zeta_b < \gamma_b$, both binned operators are unchanged. Thus their recovery protocol is unchanged for a fixed initial vector and cutoff.

Proof. For every sign string, $|a'q'_t - aq_t| \le \zeta_a$ by the triangle inequality. Under the strict margin condition no eigenvalue crosses a bin edge. The product eigenvectors are fixed, so identical signs give identical operators. □

Background: POVMs and unsharp measurements. A two-outcome measurement need not be a projection. The most general one is described by two positive operators $E_\pm$ summing to the identity (a positive-operator-valued measure, POVM); the outcome probabilities are $\langle E_\pm \rangle$ and the "observable" $E_+ - E_-$ is a Hermitian operator with eigenvalues between -1 and 1, a *contraction*, rather than an involution with eigenvalues exactly ± 1. Replacing the hard sign by a smooth function of $\cos q$ models a detector of finite resolution: outcomes near a bin edge are assigned to either bin with some probability.

For a specified unsharp version, take $F_\eta(q) = \tanh(\cos q/\eta)$, $\eta > 0$, and write $X_\eta = F_\eta(Q_r)$, $Z_\eta = F_\eta(Q_w)$. These Hermitian contractions define binary POVMs $(I \pm X_\eta)/2$ and $(I \pm Z_\eta)/2$. With

$$h_a = 1 - \tanh\left(\frac{\sin\gamma_a}{\eta}\right), \qquad a = r, b, \tag{20}$$

functional calculus gives

$$\begin{aligned} \|X_\eta - X\| \le h_r, \qquad \|Z_\eta - Z\| \le h_b, \\ \|\{X_\eta, Z_\eta\} - \{X, Z\}\| \le 2(h_r + h_b), \\ \|I - X_\eta^2\| \le 2h_r - h_r^2, \qquad \|I - Z_\eta^2\| \le 2h_b - h_b^2. \end{aligned} \tag{21}$$

Indeed, $|\cos(aq_t)| \ge \sin\gamma_a$ for every lattice point, so $|F_\eta - F| = 1 - \tanh(|\cos q|/\eta) \le h_a$ on the spectrum, which is the first line; subtracting the products $X_\eta Z_\eta - XZ$ and $Z_\eta X_\eta - ZX$, and using the contraction bounds, proves the second line; the third follows from $|F_\eta| \ge 1 - h_a$ on the spectrum. For positive margins, $h_a \le 2e^{-2\sin\gamma_a/\eta}$, whereas at an edge $F_\eta = 0$ and its difference from the hard convention is one. The bound consequently does not become small there. A small smoothed anticommutator alone is not a Pauli algebra: the unsharpness bounds must also be considered. The exact recovery below applies to the hard involutions, not unchanged to these POVMs.

5 Exact Pauli recovery and its efficiency

5.1 Reducing spectral filter and polar correction

Background: two reflections and Jordan's lemma. In finite dimensions, two involutions X and Z (think of two mirrors, or two reflections of the plane) that do not anticommute can always be brought to a simple form. Consider $K = \frac{1}{2}(XZ + ZX)$. One checks that K commutes with both X and Z: $XK = \frac{1}{2}(XXZ + XZX) = \frac{1}{2}(Z + XZX)$ and $KX = \frac{1}{2}(XZX + ZXX) = \frac{1}{2}(XZX + Z)$, using $X^2 = I$, so $XK = KX$, and similarly for Z. So the space splits into eigenspaces of K, and on each, X and Z act independently. A K-eigenspace with eigenvalue $k = \cos\theta$, $0 < |k| < 1$, decomposes further into two-dimensional blocks (as many as the eigenvalue's multiplicity, and in our problem that multiplicity is large), on each of which one can choose a basis in which $X = \sigma_x$ and $Z = \cos\theta\,\sigma_x + \sin\theta\,\sigma_z$: two reflections at an angle θ to each other. If $k = \pm 1$ then $\theta = 0$ or π, $Z = \pm X$, and the block is one-dimensional: the two measurements are the same measurement, and no second independent direction exists. If $k = 0$, $\theta = \pi/2$ and X, Z anticommute. For $0 < |k| < 1$ the pair is "almost" a Pauli pair, and one can define a corrected $\widetilde{Z} = (Z - \cos\theta\, X)/\sin\theta = \sigma_z$ that anticommutes with X exactly. This is Jordan's lemma (Halmos's two-subspace theorem [34]);

the operator identities below are its basis-free form and also apply in infinite dimensions, where a direct integral may replace the discrete block decomposition.

Let X, Z be self-adjoint involutions on a Hilbert space and set

$$K = \frac{XZ + ZX}{2}, \qquad R = Z - KX, \qquad E = \mathbf{1}_{[-\kappa,\kappa]}(K), \qquad 0 < \kappa < 1. \tag{22}$$

Here E is the projector onto the eigenvectors of K with eigenvalue in $[-\kappa, \kappa]$: a *spectral filter* that keeps the blocks on which the two reflections are at an angle θ with $|\cos\theta| \le \kappa$ and discards the rest. K commutes with X and Z; direct multiplication gives $\{X, R\} = 0$ and $R^2 = I - K^2$. The range of E reduces X, Z and K, and on it $(I - K^2)^{-1/2}$ is bounded (since $1 - k^2 \ge 1 - \kappa^2 > 0$ there). Define there

$$\widetilde{Z} = R\,(I - K^2)^{-1/2}, \qquad \widetilde{Y} = iX\widetilde{Z}. \tag{23}$$

These are self-adjoint involutions with $\{X, \widetilde{Z}\} = 0$, hence $(X, \widetilde{Y}, \widetilde{Z})$ is an exact Pauli algebra on $\operatorname{ran} E$, generally with multiplicity (many copies of a qubit, one per block). This is the Jordan two-involution reduction (Halmos's two-subspace theorem [34]). In finite dimensions, X and Z decompose into two-dimensional blocks on which they act as σ_x and $\cos\theta\,\sigma_x + \sin\theta\,\sigma_z$, with $K = \cos\theta$; the filter removes the one-dimensional blocks with $K = \pm 1$. In general, the spectral decomposition may be a direct integral, with multiplicity, rather than a discrete sum. The bounded operator identities above apply in either case and give the representation explicitly: since X and $\widetilde{Z}$ are anticommuting involutions, $\operatorname{ran} E \cong \mathbb{C}^2 \otimes \mathcal{M}$ with $X = \sigma_x \otimes I$ and $\widetilde{Z} = \sigma_z \otimes I$; since K commutes with both, $K = I \otimes C$ for a self-adjoint C on $\mathcal{M}$ with $\|C\| \le \kappa$; and then $Z = KX + R = \sigma_x \otimes C + \sigma_z \otimes \sqrt{I - C^2}$. The factor $\mathcal{M}$ is the complementary logical factor referred to in the Introduction. The name "polar correction" refers to the fact that $\widetilde{Z} = R|R|^{-1}$ is the unitary part of the polar decomposition of R. We claim no novelty for this algebraic step, which is the standard tool of self-testing [29, 30, 35]; the content here is its application to specified coboson quadratures and the calculation of the rejected weight, the measurement correction and the success probability.

The rejected weight and the correction are controlled by the original defect. For a normalised initial state v let $t = \langle v|E|v\rangle$, the probability that the filter accepts v. Since $\varepsilon_v = 2\|Kv\|$ (compare (15) and (22)), Chebyshev's inequality – the weight of v on eigenvalues $|k| > \kappa$ is at most $\|Kv\|^2/\kappa^2$ – gives

$$t \ge \max\Big(0,\, 1 - \frac{\varepsilon_v^2}{4\kappa^2}\Big). \tag{24}$$

Moreover, on the retained sector $\{\widetilde{Z}, Z\} = 2\sqrt{I - K^2}$, so

$$(\widetilde{Z} - Z)^2 = 2\big(I - \sqrt{I - K^2}\big), \qquad \|\widetilde{Z} - Z\|_E \le \sqrt{2\big(1 - \sqrt{1 - \kappa^2}\big)}, \qquad \|(\widetilde{Z} - Z)Ev\|^2 \le \varepsilon_v^2/2, \tag{25}$$

the last using $1 - \sqrt{1 - k^2} \le k^2$. A smaller cutoff tightens the operator correction but may reject more weight. It should be kept in mind that $\widetilde{Z}$ is a nonlinear function of *both* conjugate quadratures; at finite rank it is not a binned quadrature measurement, and realising it requires coherent control of a two-quadrature observable.

5.2 Singlet projector, success probability, and a mixedness criterion

Background: the singlet, the Bloch vector and purity. Two qubits have the singlet state $(|01\rangle - |10\rangle)/\sqrt{2}$, the unique state with $\sigma_i \otimes \sigma_i = -1$ for all three i. The projector onto it is $\frac{1}{4}(I - \sigma_x \otimes \sigma_x - \sigma_y \otimes \sigma_y - \sigma_z \otimes \sigma_z)$. A single qubit's state is a 2×2 density matrix $\rho = \frac{1}{2}(I + \vec{m} \cdot \vec{\sigma})$ with Bloch vector $\vec{m}$, $|\vec{m}| \le 1$; $|\vec{m}| = 1$ for a pure state and $|\vec{m}| < 1$ for a mixed one, with $\operatorname{Tr}\rho^2 = \frac{1}{2}(1 + |\vec{m}|^2)$. When a qubit is part of a larger pure state, its reduced density matrix is mixed exactly when the qubit is entangled with the rest. Finally, the singlet is antisymmetric

under exchange of the two qubits, so a symmetric product state $|\chi\rangle \otimes |\chi\rangle$ has zero overlap with it; two copies of a pure qubit can never be projected onto the singlet.

Extend the corrected operators by zero outside the retained sector and write $X_E = EXE$. The joint observables $X_E \otimes X_E$ and $\widetilde{Z} \otimes \widetilde{Z}$ commute, and their simultaneous minus-sector projector is

$$P_-^{(\kappa)} = \frac{1}{4}\big(E \otimes E - X_E \otimes X_E - \widetilde{Y} \otimes \widetilde{Y} - \widetilde{Z} \otimes \widetilde{Z}\big). \tag{26}$$

This is the singlet projector of the recovered qubit, restricted to the accepted sector on both sides. For identical normalised local initial states v, with $m_i = \langle v|\sigma_{i,E}|v\rangle$, $\sigma_{i,E} \in \{X_E, \widetilde{Y}, \widetilde{Z}\}$, idempotence gives the total probability including both local filters,

$$p(v) = \big\|P_-^{(\kappa)}(v \otimes v)\big\|^2 = \frac{1}{4}\big(t^2 - m_x^2 - m_y^2 - m_z^2\big). \tag{27}$$

For $p > 0$ every normalised projected state has $\langle \sigma_{i,A}\sigma_{j,B}\rangle = -\delta_{ij}$ and vanishing local Pauli means; with Alice's axes $X_E, \widetilde{Z}$ and Bob's $(X_E \pm \widetilde{Z})/\sqrt{2}$ the CHSH value [36] is $2\sqrt{2}$, Tsirelson's bound [37] (the largest value quantum mechanics allows), as for any exact Pauli pair in a singlet. Within the theory this value therefore carries no information about the composite; the model-dependent quantities are t, the correction (25), and p. Its role is experimental: by the self-testing theorems [30, 35], an observed CHSH value of $2\sqrt{2}$ certifies, without any model of the devices or of the composite, that the prepared state is a singlet and the measurements are a Pauli pair on its support, up to local isometries[3]; a Bell test therefore verifies the output of an implementation of the filter and polar correction, though not the internal operations that produced it. The Bell value is informative within the theory only for the *uncorrected* protocol, whose conditional state is not a singlet: there it measures, in the units of a Bell experiment, how far the binned quadratures are from a Pauli pair, and the same self-testing bounds turn it into a certified singlet fidelity (Sec. 6, Appendix A).

For $t > 0$, decompose $\operatorname{ran} E \cong \mathbb{C}^2 \otimes \mathcal{M}$ into qubit and multiplicity factors: each block is a copy of the qubit, and $\mathcal{M}$ labels the copies. The filtered pure state $Ev/\sqrt{t}$ has reduced qubit state ρ_v with Bloch vector $\vec{m}/t$, and

$$p(v) = \frac{t^2}{4}\left(1 - |\vec{m}/t|^2\right) = \frac{t^2}{2}\left(1 - \operatorname{Tr}\rho_v^2\right). \tag{28}$$

This is the two-copy identity for the antisymmetric qubit projector $(I - \mathrm{Swap})/2$, used in purity and concurrence measurements [38]. Here it separates the accepted-weight factor t^2 from the conditional success probability. For $t = 0$, $p = 0$ and the normalized filtered state is undefined. For $t > 0$, preparation succeeds if and only if ρ_v is mixed, or equivalently the filtered pure state is entangled between its qubit and multiplicity factors. A state in one Jordan block cannot succeed. Merely occupying several blocks is insufficient: $|\chi\rangle \otimes (|a\rangle + |b\rangle)/\sqrt{2}$ still has a pure reduced qubit. The rank-four spectrum $\lambda^{(a)}$ of (16) is an instance: at $r = 1$ its K equals $-1/2$ on the whole vacuum, so $t = 1$ at $\kappa = 1/2$, yet $p = 0$, because the vacuum is a product of a qubit state and a multiplicity state across four blocks of equal angle.

The factor $\mathcal{M}$ is a logical subsystem defined by the two observables – the label of the Jordan block – not a spatial or constituent subsystem; the statement that the filtered vacuum is entangled between the qubit and $\mathcal{M}$ is a statement in this factorisation, and the same structure exists for the elementary oscillator, whose vacuum at $r = 1$ has a reduced-qubit Bloch length 0.85. What is specific to the composite is how the block weights, and with them t and p, depend on the Schmidt spectrum. A composite whose filtered vacuum sits in a single block, however large its accepted weight, yields nothing. The initial-state spectral measure of K (how the weight of v is distributed over the eigenvalues of K) determines t, but not in general the three Pauli means or p. For instance,

[3] That is, there exist isometries acting on Alice's and Bob's spaces separately that map the prepared state and the measured observables onto a singlet and a Pauli pair, tensored with an uncorrelated remainder. Sec. 6 gives the quantitative version (extractability).

$X = \sigma_x \otimes I$, $Z = \sigma_z \otimes I$ give $K = 0$ and $t = 1$ for every initial state. A product state between qubit and multiplicity has $p = 0$, whereas a maximally entangled state of those factors has $p = 1/4$, with the same K spectral measure.

Lemma 1. *For fixed X, Z, v, the total probability $p_\kappa(v)$ is nondecreasing in $0 < \kappa < 1$.*

Proof. On $E_0 = \mathbf{1}_{(-1,1)}(K)$ the polar part of R defines a bounded corrected Pauli representation. Each E_κ reduces it, and the corresponding singlet projector is the restriction of the same projector P_-^0: $P_-^{(\kappa)} = (E_\kappa \otimes E_\kappa)P_-^0$. These commuting projections are nested as κ increases (a larger cutoff keeps more eigenvectors). Their expectation in $v \otimes v$ is therefore nondecreasing. □

The jump locations at finite rank come from $|K|$, but their heights depend on the initial qubit structure as well as its spectral weights. Neither a nonzero reducing sector nor positive t alone guarantees $p > 0$. Conditional CHSH saturation assumes heralded preparation before the measurement settings are chosen; for massive-particle realizations, number-changing operations would additionally require an appropriate phase reference or reservoir.

6 Flat spectrum: how fast the two detectors stop seeing the composite

For a Frenkel-like coboson the number of sites S is the only parameter, and both detectors (A) and (B) must eventually see an elementary boson, since the ladder (5) tends to the oscillator one. This section follows them up to $S = 8000$. The result, Fig. 2, is that the drive stops seeing the composite as $1/S$, while binned homodyne still sees it at $S = 8000$.

Figure 2 and Table 1 give the vacuum calculations at the natural scale $r = 1$ and cutoff $\kappa = 1/2$, for all ranks up to 400 and a selection up to 8000. Panel (a) should be read as a comparison of resolutions: both curves start at order one at $S \sim 10$, where the lattice has a handful of points per bin, but at $S = 1000$ the blue curve lies a factor of about fifty below the red one (0.0093 against 0.52). Panel (b) shows what this means for the task: the singlet probability of the corrected protocol follows the slow curve, not the fast one. The features are the following.

The two defects converge at different observed rates. The displacement defect follows the controlled asymptotic law (14); its leading term is 6.6% above the exact result at $S = 30$, 2.8% at 64, and 0.65% at 256. The hard-bin defect remains of order one at many ranks near 100 and is approximately 0.37 at $S = 8000$, where $\delta_D \simeq 0.0012$. Unweighted log–log fits over the scan (+1 edge convention) give slopes -0.146 for $64 \le S \le 8000$, -0.154 for $256 \le S \le 8000$, and -0.163 for $1024 \le S \le 8000$; these are finite-window slopes. The scatter about the fitted trend is about $\pm 15\%$ for $64 \le S \le 400$ (neighbouring ranks differ by up to 40% near $S = 100$), $\pm 6\%$ for the sampled ranks between 400 and 2000, and $\pm 5\%$ above; it narrows slowly while the trend stays shallow.

Harmonic sensitivity without a rate claim. For odd harmonic orders $u = 2m+1$, $v = 2n+1$, the expansion of the single-mode rotations suggests the leading phase-error scale

$$O\left(\frac{uv(u^2r^2 + v^2b^2)}{S}\right). \tag{29}$$

(This is the second-order term of (12) with $r \to ur$, $b \to vb$ and $\lambda = 1/S$, summed over S modes.) Comparable orders give a crossover scale $S^{1/4}$, while fixing one order gives $S^{1/3}$ for the other, with scale-dependent constants. Neither observation yields a bound on the full square-wave anticommutator, because cross terms, the nonuniform vacuum spectral measure, aliasing, and the edge projectors in (18) would all have to be controlled; in particular the weight of the square wave on harmonics beyond the crossover cannot simply be identified with the vacuum defect, so

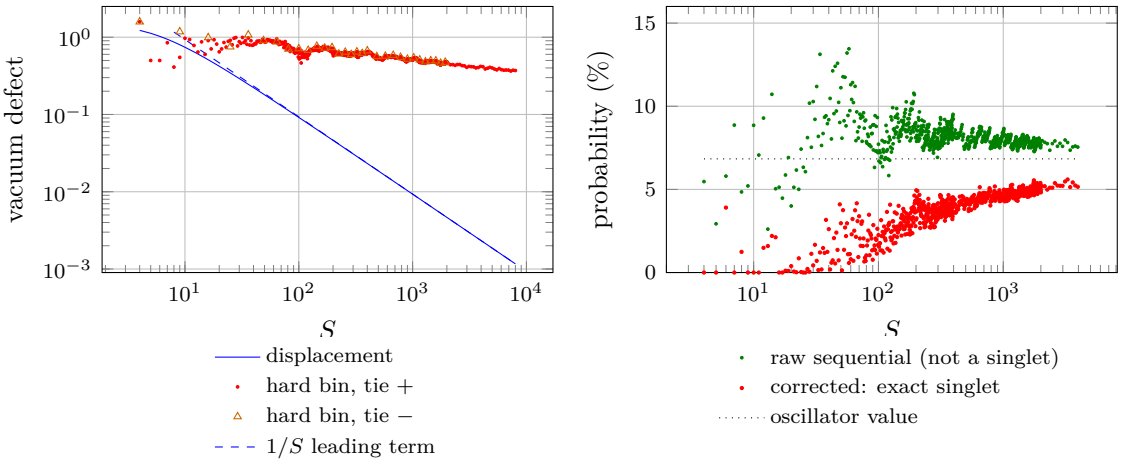


Figure 2: The two detectors on the Frenkel ladder, vacuum initial state, $r = 1$: every integer rank $4 \le S \le 400$, every fourth rank to 2000, and 30 log-spaced ranks to 8000. (a) Vacuum anticommutation defects. The drive (blue, δ_D) follows the $1/S$ law (14): it sees the composite through the commutator deviation $\Delta = 2\hat{N}/S$, which tends to zero at fixed pair number as the ladder opens. Binned homodyne (red, $\varepsilon_{\rm vac}$) sees the lattice of Fig. 1(b), whose spacing $\pi/\sqrt{S}$ shrinks slowly against the fixed bin width; its defect is still 0.37 at $S = 8000$, scatters about its trend by up to $\pm 15\%$ as lattice points cross bin edges, and at perfect squares (triangles: -1 convention; dots: $+1$) depends on the sign assigned to the points that sit exactly on an edge. (b) Probability of projecting two vacua into the singlet sector, $+1$ convention. Red: the corrected protocol of Sec. 5 ($\kappa = 1/2$, to $S = 4096$), which prepares an exact singlet; it vanishes at the ranks where the filtered vacuum has a pure reduced qubit and approaches the oscillator value $p_\infty(1) = 0.0683$ (dotted) as slowly as the red curve in (a). Green: the raw sequential protocol of Appendix A, the probability of obtaining the two "$-$" outcomes in succession; it is larger, often above the oscillator value, but since X and Z do not anticommute the conditional state is not a singlet, so this is not a success probability. All points are exact finite-rank values up to floating-point error.

we do not turn the crossover scale into a rate. Directly computed harmonic defects are given in Appendix D; they show that the fundamental-harmonic (displacement) approximation is not uniform in the harmonic order, which is the mechanism behind the slow convergence.

Exact edge atoms at perfect squares. At $r = 1$ the Q_w eigenvalue $b(S - 2j)/\sqrt{S}$ equals a bin edge $(2m + 1)\pi/2$ whenever $(S - 2j)/\sqrt{S} = 2m + 1$. If $S = n^2$ this has the integer solutions $j = (n^2 - (2m + 1)n)/2$, $2m + 1 \le n$, so every perfect-square rank has spectral atoms exactly at bin edges ($2\lceil n/2 \rceil$ of them: four at $S = 9, 16$, six at $S = 25, 36$, eight at $S = 49, 64$), while Q_r never does at $r = 1$ because $\pi/2$ is irrational. At these ranks the convention $F(\pm\pi/2) = +1$ in (15) does real work: at $S = 16$ switching from the $+1$ to the -1 convention changes t from 0.84 to 0.40 and p from 2.4×10^{-5} to 0; at $S = 9$ the alternative convention gives $t = 0$. For other scales r the same phenomenon occurs at other ranks. Any statement about a specific finite rank must therefore name the convention, and the perfect-square points in Fig. 2 should be read as convention dependent.

Preparation probability. $p_{\rm corr}$ vanishes to machine precision at the ranks $S = 4$, 5, 7, 9, 10, 11, 17 and 25, where the Bloch length in Table 1 equals 1: the filtered vacuum has a pure reduced qubit and the criterion (28) fails. It is 2.4×10^{-5} at $S = 16$ and 2.5×10^{-6} at $S = 26$, 1.2% at $S = 8$ (the exact construction of Appendix B; a favourable small rank, not a representative one), below 0.1% for most of $16 \le S \le 28$, between 0.1% and 4.2% for $32 \le S \le 127$, between 1.7% and 4.9% for $128 \le S \le 255$, and between 4.3% and 5.6% for the sampled ranks $S > 1024$; near

S	tie	δ_D	ε_0	t	$\lvert\vec{m}\rvert/t$	p_{corr}	p_{raw}	$\mathcal{B}_{\mathrm{raw}}$	edges
8	+	0.8652	0.4104	0.9698	0.9732	0.012425	0.0485	2.360	–
12	+	0.6458	0.7825	0.9531	0.9667	0.014866	0.0929	2.443	–
16	+	0.5109	0.9059	0.8436	0.9999	0.000024	0.0513	2.379	4
16	−	0.5109	0.9881	0.3988	1.0000	0.000000	0.0924	2.417	4
24	+	0.3579	0.8839	0.7756	0.9990	0.000310	0.0742	2.377	–
32	+	0.2747	0.9933	0.7433	0.9742	0.007038	0.1119	2.287	–
64	+	0.1418	0.9506	0.7504	0.9063	0.025149	0.0909	2.322	8
64	−	0.1418	0.8705	0.8122	0.9019	0.030765	0.1209	2.296	8
128	+	0.0719	0.6750	0.8392	0.9267	0.024860	0.0918	2.465	–
256	+	0.0362	0.6472	0.8670	0.9110	0.031950	0.0767	2.511	16
256	−	0.0362	0.6409	0.8652	0.9000	0.035554	0.0935	2.462	16
512	+	0.0182	0.5196	0.9047	0.8853	0.044264	0.0767	2.583	–
1024	+	0.0091	0.5404	0.8982	0.8838	0.044143	0.0765	2.572	32
1024	−	0.0091	0.5397	0.8927	0.8854	0.043058	0.0848	2.543	32
2050	+	0.0045	0.4574	0.9350	0.8732	0.051896	0.0793	2.623	–
3956	+	0.0024	0.4191	0.9422	0.8762	0.051563	0.0754	2.648	–
8000	+	0.0012	0.3707	–	–	–	0.0753	2.680	–
∞	–	0	0	1	0.8522	0.068350	0.0683	2.828	–

Table 1: Flat spectrum, vacuum initial state, $r = 1$, $\kappa = 1/2$. For every perfect-square rank shown the paired entries are for the tie-breaking conventions $F = +1$ / $F = -1$ at exact bin-edge eigenvalues; "edges" is the number of Q_w eigenvalues lying exactly on a bin edge. $\mathcal{B}_{\mathrm{raw}}$ is the Bob-optimised CHSH value of the raw sequential state (Appendix A); the corrected protocol gives $2\sqrt{2} = 2.828$ at every rank with $p_{\mathrm{corr}} > 0$. $|\vec{m}|/t$ is the Bloch length of the reduced qubit state of the filtered vacuum, Eq. (28); a value of 1 means a pure reduced qubit and hence $p = 0$. The last row is the oscillator limit of Sec. 9.

$S = 4000$ it is 5.2%, still 25% below the oscillator limit 6.8%. The raw sequential preparation has a different finite-rank success probability and does not prepare an exact singlet; both protocols approach the same oscillator limit.

The raw protocol in Bell units. The two protocols differ in what they prepare, not only in how often they succeed. The corrected protocol prepares, whenever $p > 0$, an exact singlet of an exact Pauli pair, whose CHSH value is $2\sqrt{2}$ identically. The raw sequential protocol (Appendix A: measure $Z \otimes Z$, then $X \otimes X$, keep both "$-$" outcomes) succeeds more often – p_{raw} lies between 2.6% (at $S = 13$) and 13.5% (at $S = 58$) over the ranks of Fig. 2(b), and above 6.9% for the sampled ranks $S \geq 256$ – but, because X and Z do not anticommute, its conditional state is not a singlet, and the question is how far from one it is. Figure 3 answers this in the quantity a Bell experiment would measure: the CHSH value of the raw state with Alice's settings X, Z and Bob's settings optimised (the trace-norm expression of Appendix A). It is 2.36 at $S = 8$, 2.29 at $S = 32$, 2.46 at $S = 128$, 2.58 at $S = 512$, 2.57 at $S = 1024$ and 2.68 at $S = 8000$ (Table 1); over the scan it lies between 2.09 (at $S = 17$) and 2.81, with a median that rises from 2.39 for $S < 16$ to 2.61 for $1024 \leq S \leq 4096$. It is always above the local bound 2 but still 0.15 below $2\sqrt{2}$ at $S = 8000$, and it approaches $2\sqrt{2}$ with the same slow, jittery convergence as the binned defect in Fig. 2(a), of which it is the operational expression. The Bell value can be converted into a statement about the state that does not depend on the model. Self-testing bounds give the minimum fidelity to a singlet, up to local changes of basis on each side, that any two-qubit-like state must have if it reaches a given CHSH value; the analytic bound of Ref. [30] is

$$F \;\geq\; \max\left\{\frac{1}{2},\, \frac{1}{2} + \frac{1}{2}\,\frac{\mathcal{B} - \mathcal{B}^*}{2\sqrt{2} - \mathcal{B}^*}\right\}, \qquad \mathcal{B}^* = \frac{16 + 14\sqrt{2}}{17} = 2.1058, \tag{30}$$

equal to the trivial bound 1/2 below $\mathcal{B}^*$ and to 1 at $2\sqrt{2}$ (a fidelity of 1/2 is what a product state can reach, so the bound starts there). Here "fidelity" means extractability: the fidelity to a two-qubit singlet that local isometries, allowed to discard auxiliary degrees of freedom, can attain,

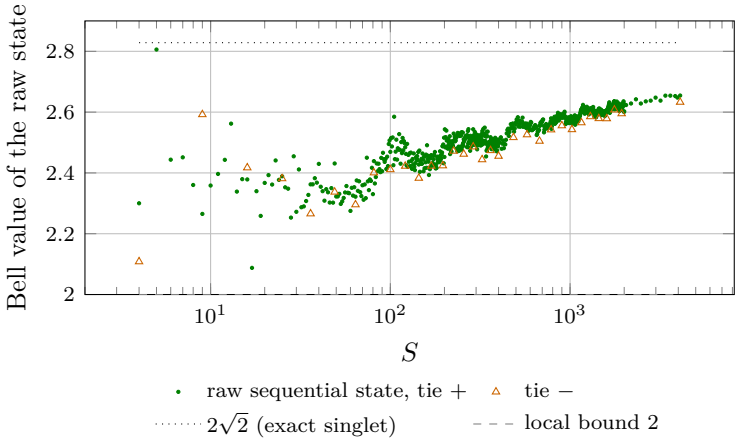


Figure 3: Bob-optimised CHSH value of the state prepared by the raw sequential protocol from two vacua, Frenkel ladder, $r = 1$, for the ranks of Fig. 2 up to 4096 (triangles: -1 convention at perfect squares). The corrected protocol gives $2\sqrt{2}$ (dotted) at every rank with $p > 0$, by construction; the raw state approaches it as slowly as the binned defect, and is still 0.26 below it at $S = 1024$ and 0.15 below it at $S = 8000$ (Table 1). The dashed line is the local-realistic bound 2; by (30), the value at $S = 1024$ certifies a singlet fidelity of at least 0.822.

not the overlap with a particular singlet vector in the original space. Applied to the raw state this gives a certified singlet fidelity of at least 0.676 at $S = 8$, 0.625 at 32, 0.748 at 128, 0.830 at 512, 0.822 at 1024, 0.874 at 3956 and 0.896 at 8000 (bounds rounded down); at $S = 17$, the worst rank of the scan, the bound gives nothing beyond the trivial value 1/2 (it is a bound, not evidence against entanglement), and for every sampled rank $S \geq 64$ with the $+1$ edge convention it is at least 0.64 (the minimum, 0.642, is at $S = 71$; with the -1 convention the $S = 64$ value is 0.631). The comparison between the two protocols is thus: a higher success probability with a state whose certified singlet fidelity lags 1 by an amount set by the quadrature lattice, against a lower success probability with a state whose singlet fidelity is exactly 1. Both converge to the oscillator values (Sec. 9).

Scale and cutoff. For the $S = 8$ vacuum the retained even-sector eigenvalues of K are 1/8 and $-3/32$, so every $\kappa \in [1/8, 1)$ gives the same probability (odd components of coherent initial states can enter at larger cutoffs), and for $\kappa < 3/32$ nothing is retained. At $S = 32$, raising κ from 0.5 to 0.9 raises the vacuum probability from 0.0070 to 0.0180. At $S = 8$ the balanced scale that minimises the leading displacement defect gives $p = 0$, while $r = 0.8$ and 1.6 give 0.0137 and 0.0264: optimising the drive (A) and optimising the homodyne task (B) are different problems.

Lattice points on bin edges are not a convention problem but a resolution problem. The two uniform sign assignments at exact edges do not coincide with the one-sided limits in the scale r, because successive edges have alternating signs; a slightly detuned homodyne phase or scale moves the edge atoms into different bins. Table 6 (Appendix E) shows this with $b = \pi/(2r)$ enforced: the $S = 8$ vacuum probability is unchanged by $\delta r = \pm 10^{-8}$, whereas at $S = 16$ it changes from 2×10^{-5} to 5×10^{-3} on either side. Proposition 1 and (21) quantify the stable regime and the limitations of smoothing at a crossing.

7 Wannier-like spectra: compositeness that does not go away

For a Frenkel-like coboson the composite nature disappears as the aggregate grows. For a Wannier exciton, a dimer, or any composite whose Schmidt distribution is localised over an infinite mode set, it does not: adding modes leaves the purity, and with it the commutator and the Weyl defect,

at a nonzero limit, and there is no oscillator limit for any detector to converge to. The question is then not how fast the composite becomes bosonic but whether a qubit can be extracted from it at all, and at what efficiency. We answer it for two model spectra. The first is the geometric class of Ref. [5],

$$\lambda_k^{(S)} = \frac{(1-z)z^{k-1}}{1-z^S}, \quad P_S = \frac{1-z}{1+z}\frac{1+z^S}{1-z^S}, \quad P_\infty = \frac{1-z}{1+z}, \qquad 0<z<1. \tag{31}$$

In Ref. [5] this spectrum arises from a two-particle Gaussian wavefunction; we use it as a model Schmidt distribution without identifying it with a specific exciton. At fixed z the purity has a nonzero limit as modes are added, so there is no oscillator limit; the limit $z \to 1$ is a different one. The omitted weight before renormalisation is z^S; since the binned observables and the spectral filter are discontinuous, a small omitted weight does not by itself bound their change.

Background: numerical methods for 2^S-dimensional spaces. With S modes the paired sector has dimension 2^S: about a million for $S = 20$. Applying Z to a vector means transforming to the product basis in which Z is diagonal (a fast transform requiring $S2^S$ operations, like a fast Fourier transform), multiplying by ± 1, and transforming back; the same for X with a different local basis. To evaluate the filter E one needs the eigenvectors of K with eigenvalue in $[-\kappa, \kappa]$. For $S \leq 12$ one can build K as a dense matrix and diagonalise it ("dense"). For larger S one uses the *Lanczos* method, which generates the space spanned by $v, Kv, K^2v, \ldots$ and diagonalises K within it; functions of K applied to v are then approximated with an accuracy that depends on how the spectrum of K is distributed near the cutoff.

The full paired space has dimension 2^S, with an invariant even-parity vacuum sector. Product Hadamard transforms apply X and Z in $O(S2^S)$ operations. For $S \leq 12$ the even sector is diagonalised densely (and, as a check, at $S = 13$ and 14 for $z = 0.7$); larger calculations use a Lanczos approximation to functions of K (Appendix E), and we keep the two regimes apart.

Checked finite-mode examples. Table 2 gives fresh dense calculations for geometric spectra and for the algebraically decaying model

$$\lambda_n^{(S)} \propto [1 + (n/w)^2]^{-4}, \qquad n = 1, \ldots, S. \tag{32}$$

The exponent is that of the hydrogenic $1s$ momentum density, but the sequence carries no three-dimensional momentum grid, angular degeneracy or normalisation volume, so it is a model of a Wannier-like tail rather than a derived exciton spectrum. Both families give finite-mode examples with $p > 0$ although the displacement and binned defects are of order one. For a sufficiently concentrated spectrum one instead obtains $X = I$: at $w = 2$, $S = 19$, $\sum_n \sqrt{\lambda_n} < \pi/2$, so all Q_r eigenvalues at $r = 1$ lie in the central bin, $K = Z$, and no sector with $|K| < 1$ remains. (All the lattice points fall into the single bin around $q = 0$, so the sign of $\cos Q_r$ is $+1$ on every eigenvector: the measurement has only one outcome and carries no information.)

Filtering and initial-state structure. For these spectra K has, besides a quasi-continuous part, eigenvalues at ± 1 and – at $z \leq 0.7$ – at $-1/2$ (to 10^{-14} in the dense calculations), with large vacuum weight: at $z = 0.5$, $S = 10$, the dense calculation gives weight 0.378 at ± 1 and 0.187 at $-1/2$; at $z = 0.7$, $S = 8$, the weight at $-1/2$ is 0.225. The weight at ± 1 sits on one-dimensional Jordan blocks, carries no qubit and must be rejected; it is the dominant loss at small Schmidt number. We have not derived the eigenvalue $-1/2$ analytically, and whether these atoms survive the infinite-mode limit is not known; we use the cutoffs 0.4 and 0.6 to stay away from it. The accepted weight is fixed by the spectral measure of K, whereas the conditional probability depends in addition on the reduced qubit state, Eq. (28).

Figure 5(a) shows the nondecreasing step functions of Lemma 1 for dense finite examples. Panel (b) gives the correction bound (25), which is 0.30 at $\kappa = 0.3$, 0.52 at 0.5 and 0.89 at 0.8. The cutoff thus sets success probability against the admissible difference between $\widetilde{Z}$ and the binned quadrature Z; $\kappa = 1/2$ elsewhere in this paper is a representative choice, not an optimised one.

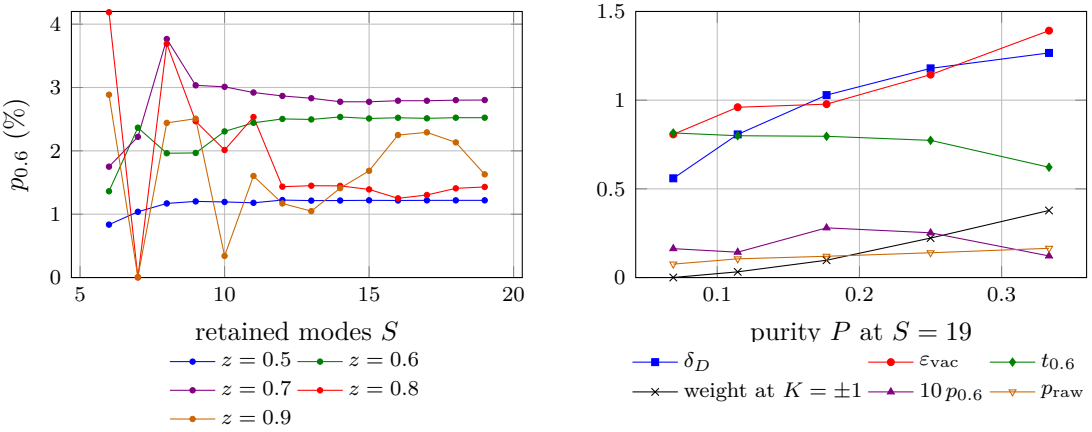


Figure 4: Wannier-like (geometric) spectra at $r = 1$: the composite stays composite as modes are added, and the singlet probability remains at the percent level over the computed truncations. Left: corrected-protocol vacuum probability at $\kappa = 0.6$ versus the number of retained modes, for Schmidt numbers $\mathcal{K}_\infty = 3$ ($z = 0.5$) to 19 ($z = 0.9$); $S \leq 12$ dense, $S \geq 13$ Lanczos. For $z \leq 0.6$ the curves change little over $13 \leq S \leq 19$; the $S = 19$ truncations have Weyl defects 1.27 and 1.18 and singlet probabilities 1.2% and 2.5%, consistent with nonzero limits (Proposition 2 and the discussion below). For $z = 0.9$ the omitted weight is still 13% at $S = 19$ and the curve has not settled. Right: the values at $S = 19$ versus the purity P (from $z = 0.9$ at the left to $z = 0.5$ at the right). As the composite becomes less bosonic both defects rise to order one, the vacuum weight on the one-dimensional Jordan blocks – which carry no qubit – rises from 0 to 0.38 and becomes the dominant loss, the accepted weight $t_{0.6}$ falls, and the singlet probability $p_{0.6}$ stays at the percent level throughout; p_{raw} is the raw sequential value, not a singlet probability. The two leftmost points ($z = 0.8$, 0.9; omitted weights 0.014 and 0.135) are finite-mode values that are still changing with S.

Behaviour with the number of modes. Figure 4 and Table 3 give the scan through $S = 19$. For $z \leq 0.6$ the quantities are stable in S once the omitted weight is small: between $S = 13$ and 19, $p_{0.6}$ varies by 4×10^{-5} (0.4% of its value, between 0.012147 and 0.012191) at $z = 0.5$ and by 4×10^{-4} (1.6%) at $z = 0.6$, $t_{0.6}$ by 6×10^{-4} in both cases, and the $\kappa = 0.4$ quantities by approximately 1–3.5%. This is finite-window stability at the percent level, not a certified value of an infinite-mode limit; the existence of that limit is discussed below. At $z = 0.7$ the dependence on S is still visible at the cutoff 0.4 – $t_{0.4}$ decreases from 0.653 at $S = 12$ to 0.603 at 19 – but not at 0.6, where $p_{0.6}$ is 0.0279 ± 0.0002 for $S \geq 14$; dense even-sector diagonalisations at $S = 13$ and 14 reproduce the Lanczos values to four digits ($t_{0.4} = 0.6245$, $p_{0.4} = 0.02072$, $t_{0.6} = 0.7838$, $p_{0.6} = 0.02831$ at $S = 13$; $t_{0.4} = 0.6474$, $p_{0.4} = 0.02156$, $t_{0.6} = 0.7770$, $p_{0.6} = 0.02775$ at $S = 14$), so the drift at $\kappa = 0.4$ is a genuine mode-truncation effect (eigenvalues of $|K|$ close to 0.4 moving across the cutoff as modes are added) and not a Krylov artefact. At $z = 0.8$ and 0.9 the omitted weights at $S = 19$ are 0.014 and 0.135, the finite Schmidt numbers $1/P_{19}$ are 8.7 and 14.5 rather than the limiting 9 and 19, and the probabilities are still changing with S ($p_{0.6} = 0.0229$, 0.0213, 0.0163 at $S = 17, 18, 19$ for $z = 0.9$); these two rows are finite-mode values only.

The Lanczos filter has its own error when the vacuum spectral measure of K is quasi-continuous near the cutoff. Table 7 (Appendix E) shows the sensitivity to the Krylov dimension at $z = 0.8$, $S = 16$: $t_{0.4}$ moves by 2% and $p_{0.4}$ by 3% between 240 and 960 steps, while the $\kappa = 0.6$ values move by less than 0.5%. Ritz values (the approximate eigenvalues produced by Lanczos) are also unreliable for locating atoms: the weight 0.08 at $-1/2$ found densely at $z = 0.7$, $S = 13$ and 14, is not resolved by the 240-step Lanczos runs at $S \geq 17$, so we make no statement about that atom at $S = 19$.

model	S	P	δ_D	ε_0	$p_{0.4}$	$p_{0.6}$
geometric $z = 0.7$	8	0.1981	1.089	0.997	0.020650	0.037659
geometric $z = 0.5$	10	0.3340	1.267	1.393	0.002998	0.011935
geometric $z = 0.7$	10	0.1867	1.058	0.978	0.020780	0.030106
algebraic $w = 3$	10	0.5153	1.303	1.631	0.000470	0.000676
algebraic $w = 4$	10	0.3813	1.302	1.513	0.001535	0.005982
algebraic $w = 6$	10	0.2480	1.189	1.156	0.005082	0.021036

Table 2: Singlet extraction despite order-one Weyl and binned defects: finite-mode vacuum examples at $r = 1$ from dense diagonalisation of the even sector, at cutoffs $\kappa = 0.4$ and 0.6. All have Weyl defects and binned defects of order one; at $\kappa = 0.6$ four of the six yield a singlet with probability above 1%, the two most concentrated algebraic spectra ($w = 3$, $\mathcal{K} = 1.9$; $w = 4$, $\mathcal{K} = 2.6$) below it.

z	$\mathcal{K}_S$	z^S	δ_D	ε_0	$t_{0.4}$	$p_{0.4}$	$t_{0.6}$	$p_{0.6}$
0.5	3.00	0.0000	1.266	1.392	0.302	0.0029	0.622	0.0122
0.6	4.00	0.0001	1.179	1.144	0.533	0.0115	0.773	0.0252
0.7	5.65	0.0011	1.029	0.978	0.603	0.0195	0.796	0.0280
0.8	8.74	0.0144	0.807	0.960	0.525	0.0072	0.800	0.0143
0.9	14.48	0.1351	0.559	0.807	0.771	0.0107	0.815	0.0163

Table 3: Geometric class at $S = 19$ retained modes, $r = 1$, vacuum initial state (Lanczos, 240 steps). $\mathcal{K}_S = 1/P_S$ is the Schmidt number of the truncated, renormalised spectrum. For $z \le 0.6$ the preparation probabilities vary by approximately 0.4–3.5% over $13 \le S \le 19$ (see text); the rows $z = 0.8$, 0.9 are finite-mode values that are still changing with S.

The infinite-mode limit: what converges. The convergence question has two parts, the bin edges of the quadratures and the filter boundaries $\pm\kappa$ of K. The first can be settled in general, for every spectrum with summable Schmidt amplitudes and for every vector, by an elementary argument.

Background: spectral measures, atoms and coin-toss sums. For a Hermitian operator Q and a normalised vector ψ, the *spectral measure* of ψ is the probability distribution of the outcomes of a measurement of Q in the state ψ. An *atom* of a distribution is a single value that carries positive probability; a distribution without atoms assigns probability zero to every individual value, although it may still be concentrated on a set of zero length (a Cantor set) rather than having a density. For the vacuum and the quadrature Q_w, the distribution is that of the random sum $-b\sum_k \sqrt{\lambda_k}\,\varepsilon_k$ with independent fair coin tosses $\varepsilon_k = \pm 1$. For the geometric class this is, up to scale, the Bernoulli convolution $\sum_n \varepsilon_n \rho^n$ with $\rho = \sqrt{z}$, a classical object: for $\rho < 1/2$ it lives on a Cantor set, for $\rho = 1/2$ it is uniform, and for $\rho > 1/2$ it has a density for almost every ρ [39, 40] but not for all [41], the complete answer being unknown. None of this finer structure is needed below: the only property used is that the distribution has no atoms, which holds for every summable sequence of amplitudes and is proved in three lines. "Strong" convergence of operators, $A_S\psi \to A\psi$ for every fixed vector, is the notion introduced in the Background paragraph of Sec. 9; it is the right notion for discontinuous functions such as sgn cos of converging operators.

Lemma 2. *Let $a_k = \sqrt{\lambda_k} > 0$ for infinitely many k, with $\sum_k a_k < \infty$, and let ε_k be independent fair signs. The law of $q = \sum_k a_k \varepsilon_k$ has no atoms.*

Proof. Since $a_k \to 0$ one can choose a subsequence $b_1 > b_2 > \dots$ of the a_k with $b_{j+1} < b_j/4$, so that $\sum_{j>m} b_j < b_m/3$. Two sign sequences on this subsequence that first differ at index m give sums differing by at least $2b_m - 2\sum_{j>m} b_j > 0$; the map from sign sequences to sums is therefore injective on the subsequence, and every value of the subsequence sum has probability zero. The

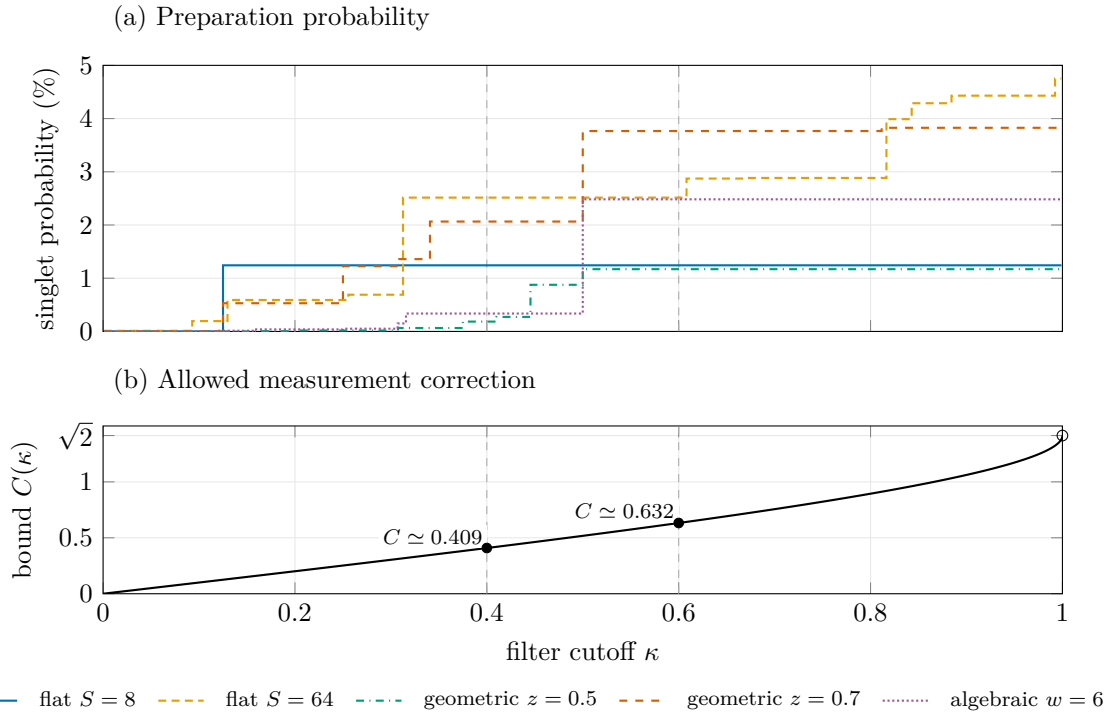


Figure 5: Success probability and allowed measurement correction as the filter is opened, at $r = 1$. (a) Total singlet probability from two vacua, including both local filters: flat spectra at $S = 8, 64$ and nonflat spectra at $S = 8$. The dense finite-rank curves are nondecreasing step functions; possible jumps occur at eigenvalues $|k|$ of $|K|$, with $k = \cos\theta$ on a Jordan block. Their heights depend on the initial state. (b) The universal bound $C(\kappa) = \sqrt{2(1 - \sqrt{1 - \kappa^2})}$ on $\|\widetilde{Z} - Z\|_E$, not the actual correction for any one model. Dashed guides mark the cutoffs 0.4 and 0.6 used in the nonflat tables: the corresponding bounds are approximately 0.409 and 0.632, respectively. The two panels share the cutoff axis, so gains in success probability can be read against the allowed change of observable without a second vertical scale. Curves are shown for $0 < \kappa < 1$; the endpoint in (b) denotes the limiting bound $\sqrt{2}$.

subsequence sum is independent of the sum over the remaining indices, and the convolution of a nonatomic law with any law is nonatomic. □

Proposition 2. *Let $(\lambda_k)_{k\geq 1}$ be a normalised infinite sequence with $\lambda_k > 0$ and $\sum_k \sqrt{\lambda_k} < \infty$, fix $r > 0$ and $b = \pi/(2r)$, and let $\lambda_k^{(S)} = \lambda_k / \sum_{j\leq S} \lambda_j$ for $1 \leq k \leq S$ be the renormalised truncations. Let $\mathcal{H}$ be the incomplete tensor product built on the vacuum[4]. Then $X_S \to X_\infty$, $Z_S \to Z_\infty$ and $K_S \to K_\infty$ strongly on $\mathcal{H}$, and for every $\kappa \in (0, 1)$ such that $\pm\kappa$ are not eigenvalues of K_∞ the filter, the corrected observables, the two-party singlet projector and the vacuum preparation probability converge to their infinite-mode values. All but countably many κ have this property.*

Proof. Expand every mode in the eigenbasis of $d_k + d_k^\dagger$ (the local σ_x). This identifies $\mathcal{H}$ with $L^2(\{-1, +1\}^{\mathbb{N}}, \mu)$, μ the fair product measure: the vacuum is the constant function 1, a finite-occupation basis vector is the Walsh character $\prod_{k\in A} \varepsilon_k$, and $Q_w^{(\infty)}$ is multiplication by $-b\, q(\varepsilon)$ with $q = \sum_k a_k \varepsilon_k$. The spectral measure of any $\psi \in \mathcal{H}$ is $\nu_\psi(A) = \int \mathbf{1}\{-bq(\varepsilon) \in A\}\, |\psi(\varepsilon)|^2\, d\mu$, hence absolutely continuous with respect to the vacuum's, which is the law of $-bq$ and has no atoms by Lemma 2; so no vector has spectral mass on the countable set of bin edges.

[4]von Neumann's incomplete tensor product: the closed span of the product vectors that differ from the vacuum product $\bigotimes_k |0_k\rangle$ in finitely many modes, i.e. the completion of the space of finite-occupation vectors. It is the natural infinite-mode state space here because the vacuum and every finite-pair state lie in it.

The truncated quadrature is multiplication by $-bq_S(\varepsilon)$ with $q_S = \sum_{k\le S} a_k^{(S)}\varepsilon_k$, and $\sup_\varepsilon |q_S - q| \le \big(\sum_k a_k\big)\big|(\sum_{j\le S}\lambda_j)^{-1/2} - 1\big| + \sum_{k>S} a_k \to 0$; off the bin edges the sign $F(-bq_S(\varepsilon))$ therefore eventually equals $F(-bq(\varepsilon))$, and dominated convergence gives $Z_S\psi \to Z_\infty\psi$ for every ψ. The quadrature Q_r equals $(r/b)\, i^{\hat N} Q_w\, i^{-\hat N}$, with $\hat N$ the pair number: conjugation by the diagonal unitary $i^{\hat N}$ of the occupation basis (a product of local phases, which preserves $\mathcal{H}$ and is independent of S) sends $B^\dagger + B$ to $i(B^\dagger - B)$. Hence the same holds for X_S.

Strong convergence of uniformly bounded operators passes to products, hence to K_S. If $\pm\kappa$ are not eigenvalues of K_∞, the spectral projector $\mathbf{1}_{[-\kappa,\kappa]}(K_S)$ converges strongly to $\mathbf{1}_{[-\kappa,\kappa]}(K_\infty)$, and the bounded Borel function $g_\kappa(k) = \mathbf{1}_{[-\kappa,\kappa]}(k)/\sqrt{1-k^2}$ (defined as zero outside the retained interval) is continuous except at $\pm\kappa$. Thus $g_\kappa(K_S)$ also converges strongly. Using $\widetilde{Z}_S = (Z_S - K_S X_S) g_\kappa(K_S)$ gives convergence of the corrected observables, their two-party singlet projectors and p. On a separable space K_∞ has at most countably many eigenvalues. □

Proposition 2 applies to both model families ($a_k \propto z^{k/2}$ and $a_n \propto (1+(n/w)^2)^{-2}$). It settles the bin-edge question for every vector and leaves the filter boundary as the remaining convergence condition. What it does not give is a proof that the particular cutoffs 0.4 and 0.6 are not eigenvalues of $|K_\infty|$ for the spectra of Table 3, a proof that the limiting probabilities are positive, or a rate. We therefore state the existence and positivity of the infinite-mode values of Table 3 as a conjecture supported by the finite-window stability reported above, and note that Proposition 2 guarantees convergence at all but countably many cutoffs. (For the geometric class the vacuum law is the Bernoulli convolution $\nu_{\sqrt{z}}$[5], whose finer properties – singularity for Pisot parameters [41], absolute continuity for almost every parameter [39, 40] and for Garsia parameters such as $z = 1/2$ [42] – are not needed here.) First binned moments are in any case insufficient to identify the algebra: at $z = 0.5$, $\langle Z\rangle_0 = 0.293$, 21% below the oscillator value, while $\langle X\rangle_0 = 0.749$ is within 3% of it.

8 Detector (C): coherence without algebra

Background: coherent states and $g^{(2)}$. For an elementary boson the coherent state $|\alpha\rangle = e^{-|\alpha|^2/2}\sum_N \alpha^N |N\rangle/\sqrt{N!}$ is the eigenstate of the annihilation operator, $a|\alpha\rangle = \alpha|\alpha\rangle$; it describes an ideal laser field. Its photon-number distribution is Poissonian and its second-order correlation $g^{(2)} = \langle a^{\dagger 2}a^2\rangle/\langle a^\dagger a\rangle^2$ equals exactly 1: detecting one photon does not change the probability of detecting another. For a thermal field $g^{(2)} = 2$ (bunching), for a single photon 0 (antibunching). Hanbury Brown–Twiss interferometry measures $g^{(2)}$ by correlating two detectors, and $g^{(2)} = 1$ is a standard signature of a coherent field – a necessary condition, satisfied also by some states that are not coherent, as Sec. 8 exploits.

Intensity correlations are widely used to assess coherence in exciton and polariton condensates [10]. A value $g^{(2)} \simeq 1$ is consistent with a coherent field, but does not by itself characterize a coherent state. The quasi-Glauber states of Ref. [3] are the composite states that pass this test best, and they let us ask whether passing it says anything about detector (B). With $\mu = |\alpha|^2$ define

$$|G_\alpha\rangle = \frac{1}{\sqrt{T_S}}\sum_{N=0}^{S}\frac{\alpha^N}{\sqrt{N!F_N}}|N\rangle, \qquad T_M = \sum_{N=0}^{M}\frac{\mu^N}{N!F_N}, \qquad \mathcal{P}_N = \frac{\mu^N}{N!F_N T_S}. \tag{33}$$

These are the coherent-state coefficients $\alpha^N/\sqrt{N!}$ corrected by the Pauli-blocking factor $F_N^{-1/2}$ so that B acts on them as closely as possible like α. The finite ladder gives

$$B|G_\alpha\rangle = \alpha\Big(|G_\alpha\rangle - \frac{\alpha^S}{\sqrt{S!F_S T_S}}|S\rangle\Big), \qquad \|(B-\alpha)|G_\alpha\rangle\|^2 = \mu\,\mathcal{P}_S, \tag{34}$$

[5]The Bernoulli convolution ν_ρ is the law of $\sum_{n\ge 0}\varepsilon_n\rho^n$ for fair signs ε_n. A Pisot number is a real algebraic integer greater than 1 whose other conjugates lie inside the unit disc; a Garsia number is an algebraic integer whose conjugates all lie outside the unit disc and whose norm is ± 2, such as $\sqrt{2}$, which is why $z = 1/2$ is covered.

S	initial state	t	$g^{(2)}$	ε_v	$p(v)$
8	vacuum	0.969828	–	0.410435	0.012424631790
8	quasi-Glauber	0.321363	0.999988135	1.571540	0.000274534936
8	Anderson	0.422886	0.789053254	1.400949	0.000000358389
32	vacuum	0.743321	–	0.993255	0.007038319914
32	quasi-Glauber	0.762299	1.000000000	0.874936	0.008913682982
32	Anderson	0.757124	0.940306901	0.880731	0.010084343692

Table 4: Initial state dependence of the corrected singlet protocol, $r = 1$, $\kappa = 1/2$, $\alpha = \beta = 1$. Entries are rounded; the field-correlation ratio is undefined for the vacuum.

so the state is an exact eigenstate of B only up to the top-rung term, which cannot vanish for finite nonzero α since $B^{S+1} = 0$ excludes a nonzero annihilation eigenvalue (a nilpotent operator has only the eigenvalue 0). Repeated annihilation yields

$$\langle B^{\dagger m} B^m \rangle_G = \mu^m \frac{T_{S-m}}{T_S}, \qquad g_G^{(2)} = \frac{T_S T_{S-2}}{T_{S-1}^2} = \frac{1 - \mathcal{P}_S - \mathcal{P}_{S-1}}{(1 - \mathcal{P}_S)^2}. \tag{35}$$

For $S = 8$, $\alpha = 1$ this gives $g_G^{(2)} = 0.9999881\ldots$ while the same state has $\varepsilon_G = 1.5715$ at $r = 1$: near-unit second-order coherence coexists with an order-one anticommutation defect of the binned quadratures. We do not read this as a surprise – $g^{(2)}$ is a property of the state, ε a property of the state and two specific discontinuous observables – but it makes concrete that (C) does not control (B).

The quasi-Glauber state must be distinguished from the displaced vacuum. For the flat spectrum and $\xi = \rho e^{i\varphi}$ the product of local rotations gives exactly

$$D_B(\xi)|0\rangle = (\cos a)^S \, e^{\beta B^\dagger}|0\rangle, \qquad a = \rho/\sqrt{S}, \quad \beta = \sqrt{S}\, e^{i\varphi} \tan a, \tag{36}$$

for $\cos a \neq 0$, i.e. the normalised Anderson state [43] of Ref. [3], Eq. (50) – the BCS-like product state $\prod_k (1 + \beta\sqrt{\lambda_k} d_k^\dagger)|0\rangle$. Its number-state coefficients are proportional to $\beta^N \sqrt{F_N/N!}$, whereas (33) contains $F_N^{-1/2}$; the distinction persists at finite rank. Appendix C derives the field correlation of the Anderson state and corrects Eq. (53) of Ref. [3].

Table 4 lists the corrected-protocol probabilities for the three initial states at $S = 8$ and $S = 32$ with $r = 1$, $\kappa = 1/2$ and both coherent-family parameters equal to one; both parity sectors are kept and the full initial normalisation is used. The rows of a group do not have equal mean pair number, so the table is not an optimisation at fixed density. At $S = 8$ the odd-parity K eigenvalues are $-1, 1, -3/4, -3/4$, so the whole odd sector is rejected and the coherent initial states lose their entire odd component; at $S = 32$ the three initial states are comparable. Exact radical expressions for the two nonvacuum $S = 8$ entries are given in Appendix B.

9 Oscillator limit and nonzero overlap

Background: convergence of operators. A sequence of operators A_S converges *strongly* to A if $A_S f \to Af$ for every fixed vector f. This is weaker than convergence in operator norm but is the natural notion when the A_S are discontinuous functions, such as $\operatorname{sgn}\cos$, of operators Q_S that converge: one cannot expect $F(Q_S)$ to be close to $F(Q_\infty)$ in norm, but one can expect it on each vector, provided the limiting operator puts no weight on the jumps of F. For unbounded operators like the oscillator quadratures the right notion is convergence of the resolvents $(Q_S + i)^{-1}$, which are bounded.

Everything above concerns finite S. We now show that for the Frenkel ladder the whole construction – binned quadratures, filter, corrected observables and singlet projector – converges to the oscillator construction of Ref. [12], and that the oscillator vacuum overlap at which it

arrives is strictly positive at every scale. Embed the symmetric ladder in $\ell^2(\mathbb{N}_0)$ and extend the finite annihilator by zero above S. The coefficients in (5) converge to the oscillator coefficients on every finite-number vector. Since these vectors form a core for each limiting quadrature, the finite quadratures converge in the strong resolvent sense[6]: for a core vector f,

$$(Q_S + i)^{-1}(Q_\infty + i)f - f = (Q_S + i)^{-1}(Q_\infty - Q_S)f \longrightarrow 0, \tag{37}$$

and the resolvent bound with density of the core image extends this to the whole space. Although F is discontinuous, its discontinuity set is countable and has zero spectral measure for the oscillator quadratures, so bounded functional calculus gives $X_S \to X_\infty$ and $Z_S \to Z_\infty$ strongly. The canonical Weyl relation at $2rb = \pi$ makes every odd harmonic of one quadrature anticommute with every odd harmonic of the other; Fejér approximants of the square wave are uniformly bounded and contain only odd harmonics, so their strong limits give $\{X_\infty, Z_\infty\} = 0$ and $K_S \to 0$ strongly. For every fixed $0 < \kappa < 1$ it follows that $E_S \to I$ and $\widetilde{Z}_S \to Z_\infty$ strongly, because $\mathbf{1}_{[-\kappa,\kappa]}(k)(1-k^2)^{-1/2}$ is bounded and continuous near the limiting spectral point $k = 0$. The two-party projectors converge strongly, so the vacuum preparation probabilities converge to the oscillator value.

Proposition 3. *Let*

$$f(s) = \frac{4}{\pi}\sum_{m=0}^{\infty}\frac{(-1)^m}{2m+1}e^{-(2m+1)^2 s^2/2}, \qquad p_\infty(r) = \frac{1 - f(r)^2 - f(\pi/2r)^2}{4}. \tag{38}$$

Then $0 < p_\infty(r) < 1/4$ *for every* $0 < r < \infty$. *In particular* $p_\infty(1) = 0.0683497058887\ldots$, *with* $f(1) = 0.767545$ *and* $f(\pi/2) = 0.370777$; *at the balanced scale* $p_\infty(\sqrt{\pi/2}) = 0.081709$.

Proof. $f(s)$ is the vacuum expectation of $F(Q)$ for a quadrature of variance s^2 (integrate the Fourier series of F term by term against a Gaussian); the series alternates with strictly decreasing terms, so $f(s) > 0$ for finite $s > 0$. For the real oscillator vacuum, complex conjugation makes the $Y = iXZ$ expectation vanish exactly. Put $x = f(r)$, $z = f(b)$ and $A = xX + zZ$; anticommutation gives $A^2 = (x^2 + z^2)I$, so the variance of A in the vacuum is $(x^2 + z^2)(1 - x^2 - z^2) \geq 0$, i.e. $x^2 + z^2 \leq 1$. If equality held, $(A - I)\psi_0 = 0$ and hence $xX\psi_0 = (I - zZ)\psi_0$. In the position representation $\psi_0(q) = \pi^{-1/4}e^{-q^2/2}$ the right-hand side has a nonzero jump at every bin edge, because $z > 0$ and the Gaussian never vanishes; the left-hand side is entire, being the Fourier transform of the Gaussian multiplied by the bounded momentum square wave. Equality almost everywhere would force equal one-sided limits at the edges, a contradiction. The upper bound follows from $x, z > 0$. □

These values were confirmed by the author of Ref. [12] (see the Acknowledgments); note that at $r = 1$ the two expectations are unequal, $f(1) \neq f(\pi/2)$. The corrected vacuum protocol therefore succeeds for all sufficiently large flat-spectrum ranks at every fixed scale and cutoff. The raw sequential state converges to the oscillator singlet as well, and its Bob-optimised CHSH value approaches $2\sqrt{2}$: compressed ideal Bob observables are Hermitian contractions whose expectation converges to this value, the trace-norm optimisation over contractions equals that over involutions, and the Tsirelson bound [37] supplies the matching upper bound. These arguments give neither a rate for the hard-bin convergence nor uniform control for occupations or amplitudes growing with S; Sec. 6 documents slow convergence over the computed range.

10 Discussion and conclusions

Whether a composite boson is bosonic depends on the detector. We have applied three detectors to one exactly solvable composite field and obtained three different answers, and we have traced the difference to one property of the composite: at finite Schmidt rank its quadrature has a discrete

[6] A core is a dense subspace on which the operator determines its closure; the Background paragraph below explains strong and strong resolvent convergence.

spectrum, the set of signed sums of the Schmidt amplitudes, which an elementary boson does not have.

A classical drive (A) is the fundamental harmonic of the quadrature and sees the composite through its commutator. Its Weyl defect is real-analytic in the Schmidt weights, is given by the product formula (11), has the dilute expansion (13) in purity and third moment, and decays as $1/S$ for a Frenkel-like coboson on S sites. Intensity correlations (C) see the state and not the quadrature; a quasi-Glauber state with $g^{(2)} = 0.99999$ has a binned anticommutation defect of order one. Binned homodyne (B) resolves the lattice: its operators are piecewise constant in the weights, so that equal-moment spectra give different defects and different singlet probabilities; they are stable only within the margin of Proposition 1, with lattice points on bin edges contributing the projectors of (18); and on the Frenkel ladder they converge to the oscillator algebra so slowly that the defect is 0.37 at $S = 8000$, where the Weyl defect is 0.001. The moment expansions of coboson theory, which control the drive, do not control the homodyne measurement; the harmonic defects of Appendix D identify the mechanism, the loss of the fundamental-harmonic approximation at orders growing with S, but we have not established a rate.

For the task of extracting a qubit, the practical conclusion is that the imperfect involutions can be repaired. The two-involution reduction, applied on the spectral sector of the anticommutator below a cutoff, gives an exact Pauli algebra and an exact singlet projector at every rank, with the rejected weight and the measurement correction bounded by the defect. The success probability from two vacua factorises into the squared accepted weight and the mixedness of the filtered reduced qubit, Eq. (28): filtering and the entanglement between the resolved qubit and its complementary logical factor play separate roles, the probability is nondecreasing in the cutoff, and the cutoff trades success probability against the admissible difference between the corrected observable and the binned quadrature. On the Frenkel ladder the construction converges strongly to that of Ref. [12], whose vacuum overlap is strictly positive at every scale (Proposition 3), so the protocol succeeds for all sufficiently large flat-spectrum ranks. For Wannier-like spectra, which never become bosonic, finite-mode calculations (dense to Schmidt number about 5, Lanczos beyond) give singlet probabilities up to 4% for Schmidt numbers between 2 and 9, the loss being dominated by the vacuum weight on the one-dimensional Jordan blocks; a composite whose spectrum is so concentrated that $\sum_k \sqrt{\lambda_k} < \pi/2$ at $r = 1$ – an explicit condition on the amplitudes, which near-unit purity alone does not imply – has $X = I$ and yields no qubit. For $z \leq 0.6$ the probabilities are stable at the percent level in the number of modes, and Proposition 2 shows that the whole construction converges in the infinite-mode limit at all but countably many cutoffs; whether the cutoffs used are among the exceptions, and whether the limits are positive, is open.

Open questions and outlook. Four problems remain open. First, a hard-bin convergence rate on the Frenkel ladder requires control of the full square-wave anticommutator, including the edge projectors of (18); the exponent -0.15 measured here is a finite-window value. Second, at fixed purity one must determine whether the chosen cutoffs are eigenvalues of $|K_\infty|$ and whether the limiting preparation probabilities are positive. Third, the step from model spectra to a specific system: a Schmidt spectrum derived from a confinement model of a quantum-dot exciton, or from a trapped pair, together with initial states of finite pair density, would say which composites carry a usable quadrature qubit and at what efficiency. Fourth, implementation requires coherent control of $\widetilde{Z}$, which depends on both conjugate quadratures, and a heralded joint preparation; the self-testing bound (30) can then certify the recovered algebra.

None of these is an obstacle to the main conclusion. A composite boson that the conventional diagnostics of bosonic behaviour place far from bosonic – a normalisation ratio F_2/F_1 a third below unity, a Weyl defect of order one, a quadrature spectrum that is visibly discrete – still yields, from its binned quadratures, an exact Pauli algebra and an exact singlet at the percent level, by a construction with an elementary-oscillator counterpart and with every loss computable from the Schmidt spectrum. At large flat rank the Weyl defect can already be small while binned quadratures still detect the finite spectral structure; near-unit intensity correlations likewise do

not guarantee a small binned defect. Seen by the right detector, compositeness is not a defect that must be removed before a composite can serve as a qubit resource; it is a structure whose consequences for that use can be computed, and used.

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A Exact rank-four calculation and the raw sequential protocol

Label the rank-four paired basis by bit strings $j = 0, \dots, 15$ (bit k of j is the occupation of mode k) and retain the even-parity integers $I = (0, 3, 5, 6, 9, 10, 12, 15)$. Let $H_{jt} = (-1)^{\mathrm{popcount}(j\,\&\,t)}/4$ be the normalised Hadamard matrix ("popcount" counts the ones in the bitwise AND), let H_e contain these eight rows, and put $D_{jj} = i^{\,\mathrm{popcount}(j)}$. With $q_t = \sum_{k=0}^{3} (-1)^{t_k} \sqrt{\lambda_k}$, at $r = 1$,

$$X = D H_e \,\mathrm{diag}[\mathrm{sgn}\cos q_t]\, H_e^T D^\dagger, \qquad Z = H_e \,\mathrm{diag}[\mathrm{sgn}\cos(\pi q_t/2)]\, H_e^T. \tag{39}$$

All entries are rational once the bin signs are fixed; for the spectra (16) the smallest distance of a q_t to a bin edge exceeds 0.02920 and 0.19352 respectively, so the signs are certified by rational square-root enclosures.

Since the uncorrected singlet polynomial need not be a projector, the raw protocol prepares the state obtained by the projective (Lüders) measurement of the two-party observable $Z \otimes Z$ followed by that of $X \otimes X$, keeping the minus result each time – joint projections onto the -1 eigenspaces, not local binned measurements followed by classical post-selection –

$$|\psi_{\mathrm{seq}}\rangle = \frac{\Pi_X^- \Pi_Z^- |00\rangle}{\sqrt{p_{\mathrm{seq}}}}, \qquad \Pi_A^- = \frac{I - A \otimes A}{2}, \tag{40}$$

the ordering being part of the protocol. For a real initial state v the unnormalised coefficient matrix is

$$M = \frac{1}{4}\big[vv^T - (Zv)(Zv)^T - (Xv)(Xv)^T + (XZv)(XZv)^T\big], \qquad p_{\mathrm{seq}} = \mathrm{Tr}(M^T M), \quad \Psi = M/\sqrt{p_{\mathrm{seq}}}. \tag{41}$$

With Alice's settings fixed at X, Z and the CHSH convention $\mathcal{C} = X \otimes (M_0 + M_1) + Z \otimes (M_0 - M_1)$ [36], optimisation over Bob's dichotomic settings gives

$$\mathcal{B}_{\mathrm{opt,B}} = \|H_+\|_1 + \|H_-\|_1, \qquad H_\pm = \Psi^T (X \pm Z)^T \Psi^*, \tag{42}$$

since each term is a linear functional $\mathrm{Tr}(MH)$ maximised by $M = \mathrm{sgn}\, H$ with any dichotomic extension on its kernel ($\|\cdot\|_1$ is the sum of the absolute eigenvalues). Exact evaluation gives

spectrum	ε_0^2	p_{seq}	$\mathcal{B}_{\mathrm{opt,B}}$
a	1	3/32	$1 + \sqrt{3} \simeq 2.73205$
b	5/2	23/128	$(2\sqrt{31} + 27\sqrt{2})/23 \simeq 2.14432$

Each $H_\pm$ has six zero eigenvalues; the remaining pairs are $\{\mp 1/2\}$, $\{\mp\sqrt{3}/2\}$ for a and $\{(-3 \mp 2\sqrt{31})/46\}$, $\{(3 \mp 27\sqrt{2})/46\}$ for b. These are protocol-dependent values, not intrinsic Bell violations attached to the spectra.

B Exact eight-mode construction

For $S = 8$, $r = 1$, in the even-number basis $(|0\rangle, |2\rangle, |4\rangle, |6\rangle, |8\rangle)$, the bin signs for the quadrature eigenvalue labels $j = 0, \ldots, 8$ are $(-,-,+,+,+,+,+,-,-)$ for X and $(-,-,-,+,+,+,-,-,-)$ for Z; no eigenvalue lies on an edge. The matrices are

$$X = \begin{pmatrix} \frac{55}{64} & \frac{5\sqrt{7}}{32} & -\frac{\sqrt{70}}{64} & -\frac{3\sqrt{7}}{32} & \frac{7}{64} \\ \frac{5\sqrt{7}}{32} & -\frac{5}{16} & \frac{7\sqrt{10}}{32} & \frac{7}{16} & -\frac{3\sqrt{7}}{32} \\ -\frac{\sqrt{70}}{64} & \frac{7\sqrt{10}}{32} & -\frac{3}{32} & \frac{7\sqrt{10}}{32} & -\frac{\sqrt{70}}{64} \\ -\frac{3\sqrt{7}}{32} & \frac{7}{16} & \frac{7\sqrt{10}}{32} & -\frac{5}{16} & \frac{5\sqrt{7}}{32} \\ \frac{7}{64} & -\frac{3\sqrt{7}}{32} & -\frac{\sqrt{70}}{64} & \frac{5\sqrt{7}}{32} & \frac{55}{64} \end{pmatrix}, \quad Z = \begin{pmatrix} \frac{27}{64} & -\frac{9\sqrt{7}}{32} & \frac{3\sqrt{70}}{64} & -\frac{\sqrt{7}}{32} & -\frac{21}{64} \\ -\frac{9\sqrt{7}}{32} & -\frac{9}{16} & -\frac{3\sqrt{10}}{32} & \frac{3}{16} & -\frac{\sqrt{7}}{32} \\ \frac{3\sqrt{70}}{64} & -\frac{3\sqrt{10}}{32} & -\frac{23}{32} & -\frac{3\sqrt{10}}{32} & \frac{3\sqrt{70}}{64} \\ -\frac{\sqrt{7}}{32} & \frac{3}{16} & -\frac{3\sqrt{10}}{32} & -\frac{9}{16} & -\frac{9\sqrt{7}}{32} \\ -\frac{21}{64} & -\frac{\sqrt{7}}{32} & \frac{3\sqrt{70}}{64} & -\frac{9\sqrt{7}}{32} & \frac{27}{64} \end{pmatrix}. \tag{43}$$

The characteristic polynomial of $K = (XZ + ZX)/2$ is $(u+1)(8u-1)^2(32u+3)^2/65536$; with $\kappa = 1/2$ the retained projectors and polar factor are

$$\begin{gathered} E_a = \tfrac{256}{63}(K + I)(K + \tfrac{3}{32}I), \qquad E_b = -\tfrac{1024}{203}(K + I)(K - \tfrac{1}{8}I), \qquad E = E_a + E_b, \\ J = \tfrac{8}{3\sqrt{7}}E_a + \tfrac{32}{\sqrt{1015}}E_b, \qquad \widetilde{Z} = (Z - KX)J. \end{gathered} \tag{44}$$

Direct multiplication gives $E^2 = E$, $\widetilde{Z}^2 = E$ and $\{EX, \widetilde{Z}\} = 0$. The vacuum entries of (27) are

$$t = \frac{225}{232}, \qquad m_x = \frac{1539}{1856}, \qquad m_y = 0, \qquad m_z = \frac{7\sqrt{1015}}{1856} + \frac{\sqrt{7}}{8}, \tag{45}$$

giving

$$p_8(|0\rangle) = \frac{959 - 49\sqrt{145}}{29696} = 0.0124246317901\ldots, \qquad \frac{|\vec{m}|}{t} = 0.9732. \tag{46}$$

Since any $\kappa \in [1/8, 1)$ retains the same even sector, (46) is independent of the cutoff in that range. The odd-parity K eigenvalues are $-1, 1, -3/4, -3/4$, so none is retained; with the full normalisations of the $\alpha = \beta = 1$ initial states,

$$p_8(G_1) = \frac{536363635311 - 37848641985\sqrt{145}}{293607926091776}, \qquad p_8(A_1) = \frac{361243454159 - 29996554273\sqrt{145}}{103543549572096}, \tag{47}$$

both positive, reproducing Table 4. The symbolic multiplication of the two-party coefficient matrix $(uu^T - xx^T - yy^T - zz^T)/4$, with $u = Ev$, $x = X_E v$, $y = \widetilde{Y}v$, $z = \widetilde{Z}v$, verifies all three singlet stabilisers.

C Field correlation versus pair-count correlation

For the normalised Anderson state $|A_\beta\rangle \propto e^{\beta B^\dagger}|0\rangle = \prod_k (1 + \beta\sqrt{\lambda_k}\, d_k^\dagger)|0\rangle$ with flat spectrum, each mode is independently occupied with probability $q = |\beta|^2/(S + |\beta|^2)$, so the pair-number law is binomial and $\langle(\hat{N})_k\rangle = (S)_k q^k$ for falling factorials $(x)_k = x(x-1)\cdots(x-k+1)$. Using (6) and the two-step ladder action,

$$\begin{aligned} I_A &= \langle B^\dagger B\rangle = Sq - (S-1)q^2, \\ J_A &= \langle B^{\dagger 2} B^2\rangle = \frac{S-1}{S}\, q^2 \big[S(S-1) - 2(S-1)(S-2)q + (S-2)(S-3)q^2\big], \end{aligned} \tag{48}$$

the second from $(\hat{N})_2(S - \hat{N} + 1)(S - \hat{N} + 2) = (\hat{N})_4 - 2(S-1)(\hat{N})_3 + S(S-1)(\hat{N})_2$. Hence, for $q > 0$,

$$g_A^{(2)} = \frac{J_A}{I_A^2} = \frac{S-1}{S} \cdot \frac{S(S-1) - 2(S-1)(S-2)q + (S-2)(S-3)q^2}{[S - (S-1)q]^2}. \tag{49}$$

This tends to $(1-1/S)^2$ as $q \to 0$ and to the saturated Fock value $2(1-1/S)$ as $q \to 1$, and it differs from the amplitude-independent value $\hat{g}_2 = 1 - 1/N_s$ given in Eq. (53) of Ref. [3] for the same state, Eq. (50) there, with the field-operator definition of Eq. (2) there. The value $1-1/N_s$ is exactly the binomial pair-count ratio $\langle \hat{N}(\hat{N}-1)\rangle/\langle \hat{N}\rangle^2$, which is distinct from the field correlation because of (6) (the field intensity $B^\dagger B$ is not the pair number); we do not infer the origin of the discrepancy from this coincidence.

Two independent checks: (i) in the dilute limit $\beta \to 0$, $B^{\dagger 2}|0\rangle = \sqrt{2(1-1/S)}|2\rangle$ gives $J_A \simeq |\beta|^4(1-1/S)^2$ and $I_A^2 \simeq |\beta|^4$, so $g_A^{(2)} \to (1-1/S)^2$ directly; (ii) for $S=2$, $\beta=1$ the normalised state is $(2/3, 2/3, 1/3)^T$, B has unit first superdiagonal, $I_A = 5/9$, $J_A = 1/9$ and $g_A^{(2)} = 9/25$ rather than $1/2$, confirmed by full two-site multiplication. In addition, (49) was verified against direct evaluation of $\langle B^{\dagger 2}B^2\rangle/\langle B^\dagger B\rangle^2$ on the normalised state $e^{\beta B^\dagger}|0\rangle$ in the full 2^S-dimensional paired space for $S = 3, 4, 5$ and $\beta = 0.5, 1, 2$, to all digits; e.g. $S=3$, $\beta=1$ gives $8/15 = 0.5333$ against $2/3$. The state is that of Eq. (50) of Ref. [3], the site product $\prod_p (1 + \alpha N_s^{-1/2} e^{i\theta_p}\alpha_p^\dagger \beta_p^\dagger)|0\rangle$, and the correlation function is that of Eq. (2) there. This correction does not affect the Frenkel ladder or the quasi-Glauber recurrence used in the main text; it does affect the statement in Ref. [3] that the Anderson state has $\hat{g}_2 < 1$ for all α, which by (49) holds for $S \geq 3$ only below a crossing density $q^*(S)$ with $g_A^{(2)}(q^*) = 1$ ($q^* = 0.61$ at $S = 8$, 0.52 at $S = 32$); for $S = 2$ it holds for all β.

D Harmonic-pair defects

For the flat spectrum, each ordered product of two local displacement factors maps the vacuum to a product of identical two-component vectors. Expanding each cosine into its two exponentials gives eight product states in the anticommutator. If their local vectors are u_a, then

$$\big\| \{\cos(uQ_r), \cos(vQ_w)\}|0\rangle \big\|^2 = \frac{1}{16} \sum_{a,b=1}^{8} \langle u_a | u_b \rangle^S. \tag{50}$$

(The overlap of two product states is the product of the S identical local overlaps.) This is independent of the diagonalisation of the Dicke ladder and agrees with it. Table 5 shows how the defect grows with the harmonic order at fixed rank: the displacement (fundamental) approximation is not uniform in the order, and at $S = 4096$ the $(m, 0)$ defect rises from 0.001 at $m = 0$ to 0.95 at $m = 8$. At $S = 256$ the Q_w grid is $q = k\pi/16$, so $\cos(17q) = (-1)^k \cos q$ and the $(0, 8)$ entry equals the $(0, 0)$ entry although the operators differ: aliasing is a further signature of the discrete spectrum.

m	n	$S = 256$	$S = 4096$
0	0	0.019289	0.001211
3	0	0.964957	0.093429
8	0	0.469799	0.947752
0	8	0.019289	0.213068
3	3	0.706428	0.853349

Table 5: Vacuum anticommutator defects $\|\{\cos(uQ_r), \cos(vQ_w)\}|0\rangle\|$ of harmonic pairs at $r = 1$, $u = 2m+1$, $v = 2n+1$, from (50). In the oscillator every entry vanishes.

E Numerical methods and verification

Flat-spectrum quadratures are evaluated in the invariant Dicke ladder with a tridiagonal eigensolver; the bin signs are assigned on the analytic eigenvalue grid $(2j - S)/\sqrt{S}$, with a tie tolerance proportional to machine precision and both tie conventions computed at perfect-square ranks. Non-flat operators use normalised fast Hadamard transforms in the product bases of the local σ_x and σ_y

operators, with the number-dependent phase for X; the implementation is tested against directly diagonalised tensor-product quadratures on complex vectors and reproduces the Dicke-ladder results for flat spectra to all digits. Dense recovery calculations use the invariant even vacuum sector (2^{S-1} dimensions, used up to $S = 14$) and record eigenpair residuals; larger calculations use a fully reorthogonalised Lanczos basis of 240 steps started at the vacuum, whose accuracy was assessed by comparison with dense results ($S = 13$ and 14, Sec. 7) and by varying the Krylov dimension (Table 7). The Lanczos filter is accurate to about 10^{-4} when κ is away from the vacuum spectral support of K and to about 1–3% when that support is quasi-continuous near κ; it does not reliably locate spectral atoms, which were therefore identified only from dense calculations. The cutoff curves of Fig. 5 use dense eigensystems and the computed eigenvalues of $|K|$ as step locations.

Two auxiliary tables are collected here. Table 6 gives the vacuum preparation probability at $S = 8$ and 16 under scale changes of $\pm 10^{-8}$ with $b = \pi/(2r)$ enforced (Sec. 6); Table 7 the dependence of the Lanczos filter on the Krylov dimension (Sec. 7).

S	r	p	t
8	$1 - 10^{-8}$	0.01242463	0.969828
8	1	0.01242463	0.969828
8	$1 + 10^{-8}$	0.01242463	0.969828
16	$1 - 10^{-8}$	0.00513391	0.839637
16	1	0.00002405	0.843640
16	$1 + 10^{-8}$	0.00448050	0.810486

Table 6: Vacuum preparation at $\kappa = 1/2$ under small changes of scale; exact edges use $F = +1$.

steps	$t_{0.4}$	$p_{0.4}$	$t_{0.6}$	$p_{0.6}$
240	0.531596	0.007379	0.737258	0.012558
480	0.537496	0.007503	0.739581	0.012625
960	0.543015	0.007610	0.737844	0.012605

Table 7: Dependence on the Krylov dimension at fixed $S = 16$, $z = 0.8$, $r = 1$.

To avoid cancellation in (11), put $u_k = \sin^2 \alpha_k$, $v_k = \sin^2 \beta_k$, and $\ell_k = 4u_k v_k(u_k + v_k - 2u_k v_k)$. Then $|g_k|^2 = 1 - \ell_k$ and

$$\delta_D^2 = -2\,\mathrm{expm1}(L_g) + 4e^{L_g}\cos^2(\theta_g/2), \quad L_g = \tfrac{1}{2}\sum_k \log(1 - \ell_k), \quad \theta_g = \sum_k \arg g_k, \tag{51}$$

with $\delta_D = \sqrt{2}$ if a factor vanishes.

Assertion-based tests compare this expression with tensor-product exponentials, the fast transforms with direct matrices, dense and Lanczos small-rank results, the rank-four examples, the eight-mode probability and Pauli projectors, cutoff nesting, the reduced-qubit identity, the Anderson field correlation, the edge mass 127/512 at $S = 16$, and the smoothing bounds. The radicals of Appendices A and B come from exact symbolic computation. The flat scan through $S = 8000$ and the nonflat scan through 19 modes are archived with the code; the values of Tables 1–3, the rank-four and eight-mode quantities, the oscillator constants, the harmonic-pair defects and the Bell values at selected ranks were reproduced by independent implementations that do not share code with the production scripts (`code/independent/`); the remaining prose statistics (window minima, maxima and medians) are read from the archived scans. No experimental data are reported.

The scripts, the archived scans, the test and verification records and the sources of both versions of this paper are available in the repository `https://github.com/itzhr1/coboson-quadrature-spectrum`, archived at Zenodo under doi:10.5281/zenodo.23960383 (which resolves to the latest version); `reproduce.sh` runs the tests, regenerates every table and figure block of the manuscripts from the archived data, and runs the independent checks.

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